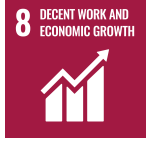




MTP Department
SIMHADRI

Note No :1



Ref:SMTTP/MAINT. PLNG./2023-24/O&M-MTP/634565

Date:07/11/2023(16:59:48)

Subject:Approval for Revised LMI on Overhaul Preparedness Index

SIMHADRI SUPERTHERMAL POWER STATION
MAINTENANCE PLANNING DEPARTMENT

Sub:- Approval of Revised LMI "LMI/OGN/OPS/SYS/45, REV-1 Dt: October 2023" on Overhaul Preparedness Index - Reg.

Operation Guidance Note (OGN) on "Overhaul Preparedness Index" was revised by COS vide **COS-ISO-00-OGN/OPS/SYST/045** in August 2023.

Based on this our station LMI on "Overhaul Preparedness Index" (LMI/OGN/OPS/SYS/45, REV-0 Dt: August 2023) is revised.

Major changes incorporated were:

1. Minimum Critical Spares required for 210/500MW KWU Turbine and Auxiliaries Overhaul were revised.
2. Minimum critical unit assemblies for Turbine and auxiliaries for ensuring availability added.
3. Relevant changes with respect to SSC done in the document. SSC replaced with USSC.
4. The validity period for Outage committee has been defined in Section 4.1.2
5. Readiness of Mechanical/ Electrical workshop before start of unit Overhauling has been included in Pre-OH checklist at Annexure-VI.

Enclosures:

1. Revised LMI on Overhaul Preparedness Index "LMI/OGN/OPS/SYS/45, REV-1 Dt: October 2023".
2. OGN on Overhaul Preparedness Index "COS-ISO-00-OGN/OPS/SYST/045".

Put up for approval by Competent Authority.

Submitted by:

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Note No:2

Few changes have been incorporated in subject LMI as per details mentioned in sl. no-3 of revision index
Approval may please be accorded for incorporating these changes.

Submitted by:

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Note No:3

LMI "Overhaul Preparedness Index" (LMI/OGN/OPS/SYS/45, REV-0) is revised as per revision in Operation Guidance Note (OGN) on "Overhaul Preparedness Index" released in September 2023.

Revised LMI is attached as annexure.

May kindly be approved.

Submitted by:

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Note No:4

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Note No:5

Review please

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Note No:6

LMI reviewed. No changes proposed.

May please process further.

Submitted by:

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Note No:7

Reviewed and put up for further approval.

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Note No:8

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On:17/11/2023(16:22:52) **Ref:**SMTTPP/MAINT. PLNG./2023-24/O&M-MTP/634565

Note No:9

LMI reviewed. No changes proposed.
May please process further.

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Note No:10

Reviewed and submitted for approval

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Note No:11

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Note No:12

May Pls. be approved

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Note No:13

please review

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On:27/11/2023(12:24:46) **Ref:**SMTPP/MAINT. PLNG./2023-24/O&M-MTP/634565

Note No:14

LMI reviewed. No changes proposed. May kindly be approved.

Submitted by:

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Note No:15

LMI reviewed.Submitted for approval.

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Note No:16

Pl review and forward with comments/changes if any

Submitted by:

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On:28/11/2023(18:04:30) **Ref:**SMTPP/MAINT. PLNG./2023-24/O&M-MTP/634565

Note No:17

Proposed LMI is reviewed and forwarded for approval.

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Note No:18

Kindly edit page no.5 as marked (revised LMI with comments, uploaded in the Annexures).

Submitted by:

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Note No:19

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Note No:20

Done as suggested and uploaded in the annexures

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Note No:21

Submitted by:

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On:12/12/2023(16:14:33) **Ref:**SMTTP/MAINT. PLNG./2023-24/O&M-MTP/634565

Note No:22

Annexure-XII in proposed LMI for Offsite is reviewed & noted.

May please be approved.

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Note No:23

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Note No:24

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NTPC Limited

SIMHADRI SUPER THERMAL POWER PROJECT

LOCATION MANAGEMENT INSTRUCTION

LMI-COS-ISO-00-OGN/OPS/SYST/045

Rev. No.: 1 October, 2023

Overhaul Preparedness Index

Approved for implementation by.....

HOP, Simhadri.

Date :

LMI REVISION APPROVAL DOCUMENTS

1. TITLE OF LMI	Overhaul Preparedness Index
2. REF. DOCUMENT CODE	COS-ISO-00-OGN/OPS/SYST/045
3. LMI REVISION NO.& DATE	REV-1 ; October'2023
4. LMI REVIEW DATE	-
5. PREPARED BY:	SRIVENKATA PHANI KUMAR CH, Sr. Manager (MTP)
6. CHECKED BY	P NAGA SURESH, DGM (MTP) RAJESH CHAWLA, HOD (MTP) SHAMBHU KUMAR SUMAN, HOD (MM-TMD) G H RAO, HOD(C&I) A.K. MUKHOPADHYAY, HOD(OFS) DHARMENDRA KUMAR TOMAR (EMD) OM PRAKASH, HOD(BMD) NILANJAN MANDAL, HOD(AHP) UMA SHANKER VERMA, HOD(EEMG) DEBJIT KUMAR KAR, HOD (C&M-STORES) A.P NAIR, AGM (TS) T R RAJA GOPALAN, GM(OPN)
7. REVIEWED BY:	MATHEW EIPE KOVOOR, GM (M) SIGNATURE:
8. REVIEWED AND APPROVED BY	SANJAY KUMAR SINHA, HOP-SIMHADRI SIGNATURE:

NTPC	LMI	SIMHADRI
TITLE	OVERHAUL PREPAREDNESS INDEX	
REF NO	LMI/OGN/OPS/SYS/45, REV-1 Dt: October 2023	

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Overhaul Preparedness Index

1.0 INTRODUCTION

The introduction of concepts like Zero forced outage by design, increasing the gap between overhauls, improving plant efficiency to design levels and attaining 105% capability of machine is the need of the hour.

The above goals can be achieved by quantum improvement in Quality of Overhauling, which requires advance planning and monitoring their preparedness levels prior to the start of the overhaul.

Preparation of Engineering declaration, quality check lists, review of scope of work and ensuring timely availability of all the materials and contracts are the vital inputs for ensuring a quality overhaul.

OPI is devised as a useful tool to measure the degree of overhauling preparedness and for taking timely remedial actions.

Formats for risk critical equipment (Risk plot and risk grid) and net heat rate improvement provided by CenPEEP are also included in this revised Engineering Declaration format of the OPI document.

2.0 SUPERSEDED DOCUMENTS: LMI/OGN/OPS/SYST/045 Rev. No.: 0 March. 2023.

3.0 REVISION DETAILS: Version 1.

- 3.1. Minimum Critical Spares required for 210/500MW KWU Turbine and Auxiliaries Overhaul were revised.
- 3.2. Minimum critical unit assemblies for Turbine and auxiliaries for ensuring availability added.
- 3.3. Relevant changes with respect to SSC done in the document. SSC replaced with USSC.
- 3.4. The validity period for Outage committee has been defined in Section 4.1.2
- 3.5. Readiness of Mechanical/ Electrical workshop before start of unit Overhauling has been included in Pre-OH checklist at Annexure-VI.

4.0 OPI MANUAL

The manual on Overhaul Preparedness Index (OPI) has been covered in detail in this LMI.

4.1 Outage Manager & Outage Committee

Head of maintenance will act as Outage Manager. MTP Head with his Long-term Planning (LTP) group will assist him exclusively for the outage management process. Outage Manager & outage committee members should be exclusively involved in planning for execution & up to completion of overhaul activities. Outage manager along with outage committee members one each from BMD, TMD, EMD, C&I, Operation, EEMG, MTP, USSC/Finance, USSC/C&M, FQA & Safety & any other person deemed relevant would form an Outage Committee

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4.1.1 Outage Manager: Roles & Responsibilities:

- ◆ He would be responsible for planning of Overhaul of all the units at station in totality and will be the focal point i.e., from proposal stage (Engg. Decl. preparation) to the successful completion of Overhaul
- ◆ He will plan O/H of the units and prepare PERT network/Activity Bar Chart in association/ involvement with LTP group. Just before start of OH, various overhauling management teams, Re-commissioning team and safety team etc. are to be formed as per station requirements.
- ◆ Engineering Declaration R0, after approval of site management, is to be created in SAP and would be submitted to RHQ-OS/COS by the outage manager.
- ◆ He will co-ordinate with concerned maintenance departments including, USSC/C&M for identification of required spares, consumables, T&Ps, Contracts etc. and their procurement/award so that all the resources are in position at least 3 months before the start of overhaul and pre overhaul checklists are filled by respective sections. Protocols /Quality check to be prepared before start of overhaul & must be completed before unit light up.
- ◆ He would co-ordinate / review the OPI review Meeting. Area maintenance engineers would continue to raise material indents and contract proposals.
- ◆ Protocols / Quality check to be prepared before start of overhaul and must be completed before unit light up.

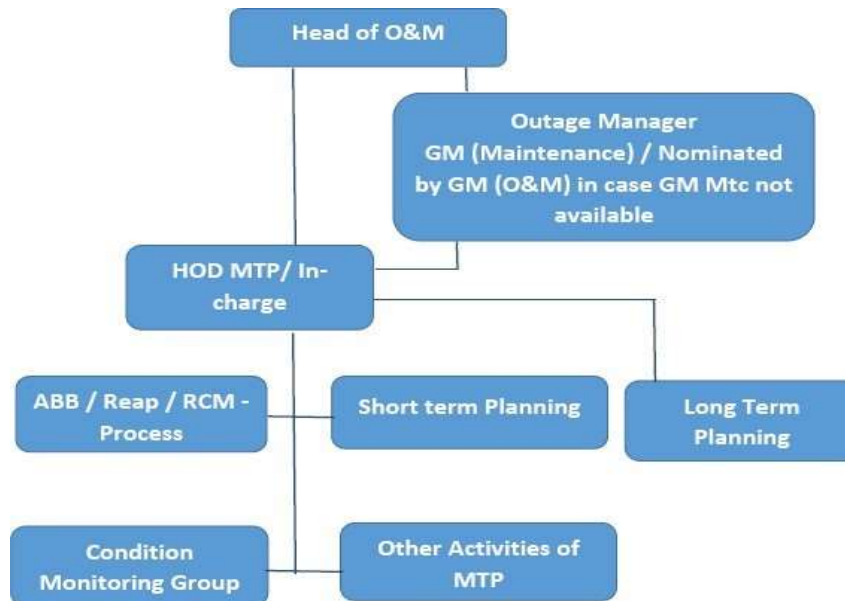
4.1.2 Outage Committee: Roles & Responsibilities:

Outage committee members drawn from different department will continue to report to their respective departmental head & coordinate & assist the outage manager for all activities related to overhauls. The validity period of Outage committee shall be two years. In case of transfer/superannuation of any committee members, another member may be nominated by the respective HOD.

- ◆ Preparation and Revision of Engineering Declaration in co-ordination with other departments. Concerned departments will provide the required inputs to outage manager through their nominated outage committee members.
- ◆ O/H Execution would be carried out by area maintenance executives. However, outage committee members would co-ordinate with various departments for timely execution of O/H.
- ◆ Outage Preparation Index (OPI) would be filled up by outage committee.
- ◆ Outage committee would generate exception report for site GM and O&M Head and would continue to co-ordinate with ROS & COS.
- ◆ Outage Manager / committee would bring out O/H Completion Report and Executive Summary of O/H including learning's in co-ordination with all concerned departments.

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4.2 New Structure of Outage Manager & Maintenance Planning



- 4.2.1 Engineering Declaration for O/H of each unit will be finalized by Outage manager/Outage Committee in the given format at least 24 months before the O/H due date. This will be done in consultation and cooperation with Area Maint. Engineers (AME) and MTP and the same is to be uploaded in SAP.
- 4.2.2 This will be revised every 6 months by Outage Manager as per the illustrative sample calendar for a typical station and will be frozen 12 Months before start of overhaul.
- 4.2.3 For convenience of review and indenting etc., Engineering Declaration Rev-0 will be made ready for each unit 24 months before the overhaul planned.
- 4.2.4 A copy of the Engg Declaration Rev-0 will be submitted by outage manager to RHQ-OS and CC-OS/MMSD 24 months before overhaul for review/comments/advice.
- 4.2.5 Revised Engg Declaration (Rev-1 onwards) need not be submitted in toto to RHQ-OS and CC-OS. Instead, a brief of the revision may be submitted such as statement of revision, changes, addition/deletion in the Scope of Work (SOW) and the reasons thereof and comments about incorporation of recommendations/remarks of KTs/RHQ-OS/CC-OS. However, it is essential that updated Engg Declaration in toto is kept handy at site by outage manager for each unit at the station.
- 4.2.6 RHQ-OS & CC-OS would review the status of Engg Declaration in May/June and Nov/Dec along with the OPI fallen due. Thereafter, review may be done/continued at the level of Station Head/O&M Head.
- 4.2.7 For illustration, the calendar for Engg Declaration and OPI has been shown in Annexure-II.
- 4.2.8 Sample OPI has been calculated for illustration in Annexure-IV.
- 4.2.9 For Gas Station, Engg Declaration for only O/H of type Major Inspection & HGPI

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i.e. (C & B), ST O/H and Generator O/H need to be submitted to RHQ- OS & CC-OS for comments/review/advice. However, Engg Declaration for minor inspection (A) also will be prepared by the station and reviewed at the station level itself. SOW resulting from the observation during such minor inspection is to be included in Major Insp & HGPI etc. and accordingly mentioned in the Engg Declaration.

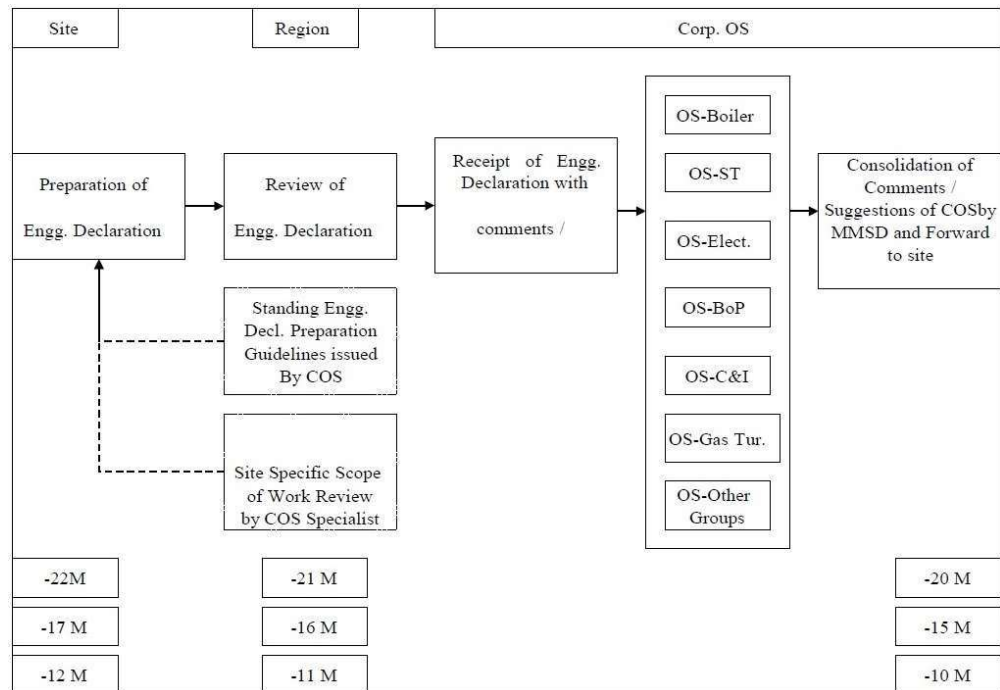
- 4.2.10** An illustrative Flow Chart for Preparation of Engg Declaration and Review of Engg Declaration is enclosed for ready reference.

4.3 Guidelines for Spares/Consumables/T&Ps /Contracts & OPI

- 4.3.1** Total quantity expected to be procured for overhaul of each unit will be estimated by Outage Manager/Outage Committee in co-ordination with Area Maintenance Engineers (AME) and the same is to be uploaded in SAP for OPI.
- 4.3.2** Outage manager will fix up the cumulative target for each month i.e., starting from (-) 24M to (-) 03M in co-ordination with Area Maintenance Engineers (AME) and USSC. So that materials/contracts are in place **3 months before the start of planned overhaul.**
- 4.3.3** Total Quantity and Cumulative Monthly Targets for each unit will be filled up in target matrix at the time of preparing Engg Declaration.
- 4.3.4** Revisions of this Target Matrix including Total Quantity will be done normally along with revision of Engg Declaration of the unit i.e., at (-)18M & (-)12M.
- 4.3.5** Revisions in between may also be incorporated based on need.
- 4.3.6** Reasons may be recorded for revising the total quantity and targets.
- 4.3.7** The site outage committee in co-ordination with outage manager will fill up the OPI Score wrt the above Target matrix and calculate the Score. For illustration, sample of OPI (-)24 has been filled up and score calculated.
- 4.3.8** The allotted Score will be reviewed in ORT or by ORT Co-Ordinator every month. In addition, status of list of Critical Items (Spares/Cons/T&Ps/Contracts) for Overhaul of each unit will be prepared and updated every month by Outage Manager and will be discussed in ORT.
- 4.3.9** ORT Exception Report will include OPI Summary Sheet and Critical Items status.
- 4.3.10** OPI review for the region shall be done by ROS with participation from CC- OS.
- 4.3.11** Deletion of any OPI items if required shall be done by the station itself. HOD MTP shall be authorized for the same.
- 4.3.12** OPI shall also be reviewed during the technical audit of the station.

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4.4 Engineering Declaration Flow chart



4.5 Guidelines for Overhauling Preparedness Index (OPI)

4.5.1 Scoring System:

Preparation for overhaul has to begin 24 months in advance. Overhaul preparation over 24 months period will be monitored through Outage Preparedness Index (OPI). There are two parts in OPI: Part A and Part B.

4.5.2 OPI Score Part A:

Monitors preparation and submission of engineering declarations and its various components. Additionally, it will also monitor other activities like incorporation of knowledge team recommendations in Scope of Work (SOW), pre-outage surveys, quality check plans, re-commissioning plans etc. These activities have to be completed by a defined milestone as indicated in the table enclosed. The table also indicated the maximum score for an activity indicated and applicable milestones. **“” has been given wherever scoring is applicable.** Thus, maximum scores applicable at 24M, 18M, 12M, 06M milestones are 200, 110, 110 and 80. Part A scores would be calculated at these milestones (24M, 18M, 12M, 06M; M stand for month; 18M means 18 months before commencement of overhaul). A typical score at 24M milestone indicated in the table in the last column. The total score achieved is 180. Hence OPI Part A score would be $180 \times 100 / 200 = 90$. Exceptions in the score would be monitored through Action Item List and Monthly exception reports (Details of the Action Item list and Monthly Exception Report is in different section). At 18M milestone, score will be calculated with respect to 18 M milestone only ie. score would be calculated based on total score of 110. Thus, if the score obtained is 80 out of 110, then OPI Part B score for 18M will be $80 \times 100 / 110 = 73\%$. Similarly, score will be calculated for 12M and 6M milestone. Thus, for an overhaul, we will have OPI Part A score as per following:

OPI	24M OPI	18M OPI	12M OPI	06M OPI
Part A Score (%)	90	73	85	95

(OPI Part B score calculation explained separately)

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PART-A

Station/Unit: O/H Type: Plan Date: Fin Year:

SI No	Activity/Description	Max	Months Before OH				OPI-24 Remarks
		Marks	24	18	12	6	Score
1	Formation of Outage Committee	20	20				Formed 20
2	Engineering Declaration Submission in required format (Revision No: --) Overhaul Date (Proposed)	30	R0 - 20	R1 - 5	R2 - 5		Submitted 20
3	Outage period & duration: Justification of outage duration & time period	30	20	5	5		Provided 20
4	Identification of Equipment & Scope of Work for Overhaul. Standard Scope of Work: 1. Boiler & Aux 2. Steam turbine & Aux 3. Turbo generator & electrical 4. Control & Instrumentation 5. Miscellaneous 6. Gas turbine & Aux	70	60	5	5		Identified 60
5	Plant Betterment Action Plan (AI/RI/EI/TO/LF) Examples: • BTL reduction plan • APH efficiency revival plan • Turbine efficiency plan • Obsolescence (R&M for DAS)	50	30	10	10		Provided but R&M not covered 20;
6	Defects/Shutdown/Pending PM Jobs included in Scope. Unit walk down observations Valve passing surveys	20				20	NA
7	Incorporation of recommendations/commence of KT's/COS/RHQ-OS	30	10	10	10		Done 10
8	Resource Analysis - Spares & Material Identification - Contracts Identification -Procurement/Award Plan	50	30	10	10		Part done 30
9	Estimated Budget Requirement	30	10	10	10		Not done 0
10	PERT & Bar Charts	30		10	10	10	NA
11	QCP Readiness - Number of QCP Identified - Number of QCP Prepared (%)	50		30	10	10	NA
12	Isolation Plan readiness checking of tools	20				20	NA
13	Re-commissioning Check Sheets (RCS) Readiness: - Number of RCS identified. -Number of RCS prepared (%)	20			10	10	NA
14	OPI Quality / Checklist Compliance	50				50	
	Total of Applicable activities	500	200	95	85	120	180

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4.5.3. Guidelines for calculation of scores for OPI Part-B:

Part B of OPI score relates to resource mobilization (spares, consumables, T&P and work contracts availability) for the overhaul.

Part B is to be created in SAP unit wise. Proper & sufficient Identification of material and services is to be done as envisaged in engineering declaration so that OPI of a particular unit should reflect the unit overhauling preparedness in real terms.

Scores for OPI Part-B would be calculated in Percentage (%) for each milestone.

‘OPI Part B SAMPLE CALCULATION’ Sheet in MS Excel (Target Matrix and calculation) is attached wherein sample OPI calculation of OPI -24, 23, 22, 21, 18, 12, 9, 6 and OPI 3 have been shown. Additionally, targets for all major milestones fixed at 21M milestone have also been indicated. **The scores for each item would be calculated based on the target for that month and actual achieved.** The guideline for setting up the targets for material procurement and works contract award for an overhaul is as per following:

For spares & special consumables having more than 12 months lead time

– Based on Engg Declaration – (R0)

Sl No	Activities Description	Target	Responsibility
1	Identification of Items	Till OPI 22	O&M
2	Purchase Indent raised & Controlled	Till OPI 21	O&M / PPG / USSC-C&M
3	Purchase enquiry issued	Till OPI 19	USSC/C&M/Fin
4	Purchase order placed	Till OPI 16	PPG / USSC / C&M / Fin
5	Items received	Till OPI 03	Site Stores

For spares and other consumables having less than 12 months lead time – Based on Engg Declaration – (R0)

Sl No.	Activities description	Target	Responsibility
1	Identification of items	Till OPI 22	O&M
2	Purchase indent raised & controlled	Till OPI 16	O&M / PPG / USSC-C&M
3	Purchase enquiry issued	Till OPI 14	USSC/C&M/Fin
4	Purchase order placed	Till OPI 11	PPG / USSC / C&M / Fin
5	Items received	Till OPI 03	Site Stores

For Spares based on Engineering Declaration –(R1)

Sl No.	Activities description	Target	Responsibility
1	Identification of items	Till OPI 17	O&M
2	Purchase indent raised & controlled	Till OPI 16	O&M / PPG / USSC-C&M
3	Purchase enquiry issued	Till OPI 15	USSC/C&M/Fin
4	Purchase order placed	Till OPI 13	PPG / USSC / C&M / Fin
5	Items received	Till OPI 03	Site Stores

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For Spares based on Engineering Declaration –(R2)

Sl No.	Activities description	Target	Responsibility
1	Identification of items	Till OPI 12	O&M
2	Purchase indent raised & controlled	Till OPI 11	O&M / PPG / USSC-C&M
3	Purchase enquiry issued	Till OPI 10	USSC/C&M/Fin
4	Purchase order placed	Till OPI 09	PPG / USSC / C&M / Fin
5	Items received	Till OPI 03	Site Stores

(Under Engineering Declaration R1 & R2 spare requirements, items are less and are to be procured on urgent basis)

Sl No.	Activities description	Target	Responsibility
1	Identification of services	Till OPI 12	O&M
2	Service proposal initiation	Till OPI 11	O&M / PPG / USSC-C&M
3	Issue of Tender	Till OPI 09	USSC/PPG/Fin
4	Issue of LOA	Till OPI 03	O&M/PPG / USSC- C&M
5	Items received	Till OPI 03	Site Stores

(All spares, consumables, T&P and contracts should be ready and available at station by 3 months before the planned overhaul date)

The targets would be set for major milestones namely 21M, 18M, 15M, 12M, 9M, 6M and 3M in the target matrix in the beginning (at 21M) as items are identified for the overhaul. Since identification of items may be carried out up to 22M based on Engineering Declaration R0, the targets would be revised up to 21M. Subsequently, after revision of Engineering Declaration R1 and R2, further identification of material would take place and accordingly, targets would be revised in 16M and 11M OPI. All target setting and calculations in Part-B would be done on the basis of no. of items (instead of no. of indents). Targets for intermediate milestones other than major milestones would be set one month before. Thus, during 20M review of an overhaul, target for 19M would be set, however targets for 18M, 15M, 12M, 9M, 6M and 3M milestones would be set at 21M.

Percentage (%) score in OPI would be calculated based on the total score of 1000 (maximum score or rated score) for each milestone.

The total score is broken up under following heads:

Spares	500 (Critical-300 & non-critical spares – 200)
Consumables	150
T & P	100
Services	250

If any of the above heads are not required to be procured, scores obtained for that head would be maximum score for that head. For example, if no T&P procurement is envisaged for an overhaul out of the required list of T&P, then score obtained for that portion of T&P would be 100 % in all OPI scores of that unit. (Since the identified items are available and are contributing towards preparedness)

Spares are further divided in 2 categories (See appendix – A for critical spares criteria)

1. Critical spares
2. Non-critical spares

Marks for Critical spares: 300

Marks for non-critical spares: 200

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Guidelines for considering a particular spare as a critical spare is covered

The following 5 activities in the process of procurement of spares, consumables & T&P have been considered for scores calculations.

1. Identification of items like spares/consumables/T&P in numbers.
2. Purchase indents raised (of number of items)
3. Purchase enquiry issued (of number of items)
4. Purchase order placed (of number of items)
5. Received of items (number of items)

Weightage for above 5 activities has been assigned as per following:

Sl No.	Description	Weightage	Marks for critical spares	Marks for Non-Critical spares
1	Identification	05%	15	10
2	Indent raised	10%	30	20
3	Enquiry issued	10%	30	20
4	PO placed	25%	75	50
5	Material received	50%	150	100
	Total	100%	300	200

Similarly, for Services weightage are given as
Services – 250

- a. Weightage for different activities for services will be as follows:

Sl No.	Description	Weightage	Scores for critical services
1	Identification of contract	05%	12.5
2	Proposal Initiation	20%	50
3	Tender Issed	15%	37.5
4	Offer received	10%	25
5	LOA issued	50%	125
	Total	100%	250

Basic guidelines for identification of critical services are as follows: -

All services pertain to Boiler, TG and its aux, electrical, C&I, AHM and Civil area directly linked to unit shut down.

In some of the cases it is observed that High value spares are going to OEM site or any other workplace for refurbishment. It needs to be available on time at our station for the commencing of quality overhaul. In this revised document the refurbishment cases have been considered.

For calculation of OPI, individual scores of each item (not percentage) is calculated and then added. Finally, OPI score, which is percentage of sum of all the individual scores and the total

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score is calculated. Normally, total score would be 1000. However, in some cases as mentioned below, the total score may be less than 1000.

If for a particular item, cumulative target fixed in a month is zero, then the total score of that item would be taken as zero and thus the total score of that section would be accordingly reduced and, in that case, total score of milestone would also be reduced. For example, at 12 OPI, the target for the no of items received (critical spares) as per target matrix is zero, then the total score for material received for the OPI would be zero and accordingly total score for the OPI would be reduced by 150. Monthly OPI score would be calculated based on total score of 850 only instead of 1000 (considering that there is no other item for which cumulative monthly target is zero).

There would be two scores: Monthly OPI Score (ie. for each OPI with respect to target matrix) and Monthly cumulative OPI Score (ie. with respect to total requirement). For example, there are 100 no. of items to be procured for an overhaul. As per target matrix fixed, PO for 60 items (critical spares) are to be placed by OPI 14M. Against this target, PO for 50 items has been placed. Total marks for PO placement is 75 for critical spares. Hence scores for PO for monthly OPI calculation would be: $(50 \times 75 / 60 = 62.5)$ and score for Cumulative OPI calculation would be: $(50 \times 75 / 100 = 37.5)$. Similarly, scores for each item would be worked out and added.

Monthly OPI Score would be calculated on the basis of cumulative target for that particular month (Refer section on Target Matrix) like 18M, 12M etc.

Cumulative scores would be calculated on the basis of total quantity required for the overhaul. A typical OPI calculation sheet in MS Excel is enclosed.

If cumulative target for a particular activity in an OPI is zero and cumulative achieved is also zero, in that case total score of that activity and actual score for that activity would be zero.

If cumulative target for a particular activity in an OPI is zero but cumulative achieved is not zero i.e. activity has been completed before milestone, in that case total score and actual score for that activity may be considered as zero.

There would be two scores: –M OPI and –M Cumulative OPI; M stands for month. Thus, there would be 15M OPI and 15M Cumulative OPI. **The former reflects the material procurement status 15 month before the overhaul with respect to the cumulative target fixed till 15 month before overhaul and the later reflects the material procurement status with respect to total requirement (material available at site / contracts LOA placed).** The ideal score of each monthly OPI would be 100 and any score less than 100 reflects shortfall in the overhaul preparation with respect to target matrix. **Thus, a score of 15M OPI of 80 reflects a shortfall in the overhaul preparation (some identified indents may not have been raised or enquiry may not have been issued by C&M).** The cumulative OPI reflects how overhaul preparation is progressing. The Cumulative OPI shall start from a small value at 24M (may be 5%) and should reach 100% at 3M OPI. It may be noted that at 3M OPI milestone, both 3M OPI score and cumulative OPI score would be equal and should be ideally 100 or very close to it as the target at this milestone would be total requirement for the overhaul.

All efforts have to be made to make OPI scores at each milestone close to 100. The gap between 100 and OPI score at a milestone is an exception with respect to the target matrix that needs to be addressed immediately.

4.6 Criteria for Critical items

1. Major replacement items specifically planned during this overhaul.
2. All items required for box up of equipment needed for 100%-unit capacity & efficiency.
3. All Capital spares items required for overhaul.
4. Long lead time items – imported and indigenous

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5. Items found damaged as surprises during overhaul.
6. All Bearings / Special T&P / Electronic cards / Special chemicals & consumables.
7. Any site-specific items.

4.7 OPI Review

4.7.1 Objective

To document and track those activities which are not completed by any particular OPI review date. This will help in overcoming the shortfalls observed during any OPI review and ensure the actions needed.

4.7.2 Review

1. Fortnightly review of OPI is done by Head of O&M with concerned USSC.
2. OPI exceptions are to be discussed in detail in ORT meeting.
3. As per the present structure of USSC C&M and Finance, OPI review meeting shall be done every quarter by ROS with outage committee members.
 - a. Status of Engineering declaration (R0, R1 & R2 etc. have been uploaded to the system or not may be reviewed.) (part A)
 - b. All the review of OPI must be strictly based on the PRs uploaded in the SAP system only. This means any PR items for discussion/review must be the part of OPI in SAP. This will ensure the proper Identification of material & services and will represent the real Overhauling preparedness of the unit. (Part B)

Six Monthly review of OPI by concerned ROS with COS and Quarterly review meeting by ROS based on SAP data.

4.8 Guidelines for OPI Monthly Exception Report (OER)

The objective of **OPI Monthly Exception Report (OER)** is to bring out the exceptions in the process of overhaul preparations and submit the same to the station management.

OER is to be prepared by Outage Manager every month after Monthly Outage Review Meeting for all overhauls. Typically, at any point of time, preparations of overhauls of three financial years are in progress (for example, current FY is 2006-07; preparations of overhauls of 2007-08 and 2008-09 are in progress). Thus, in each month, OERs of three FYs including current FY will be prepared.

There are six sections in the OER.

Section A: Resource (Material and Contract) Exceptions

This is about the exceptions of procurement processes (indent raising, issue of purchase enquiry, PO placement and receipt of material) and is to be prepared with respect to Target Matrix (Refer OPI Preparation Guideline). Exceptions in activities which were to be completed by current OPI will be brought out in this exception under “No. of Items / No. of Work Contracts” column of OER. For example, let us consider PO Placement row in the OER, current OPI is -10M for a unit, PO of 70 items were placed against the 80 items as per Target Matrix, then shortfall of 10 items will appear in the No. of Items column for PO placement in OER. PO due in last OPI and still not

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complete will come under “Pending since previous OPI column” of OER. Similarly, PO pending before last OPI and still incomplete will come under “Pending before previous OPI column”. For example, at OPI 11, there was a shortfall of 8 items out of which PO of only 3 items have been placed between OPI 10 and OPI 11. Then at OPI 10, under Pending since last OPI column, no. of items would be 3. Similarly, at OPI 10, there are 5 items whose PO is pending before OPI 11 and is still pending. Hence, under the Pending before last OPI column, there would be 5 items. In this manner, no of items for all fields namely indent raising, issue of enquiry and PO placement for spares / materials and contracts will be filled in. overall, this table is reflecting total no of shortfall under an activity and age of the shortfall. In this section, shortfall of overhaul preparation of all units will be prepared and combined. Thus, there would be an OPI 10 for unit 1, OPI 7 for unit 2, OPI 5 for unit 3 and OPI 1 for unit 4. Total shortfall for each unit under each head (indent raising, issue of enquiry etc.) will be worked out, combined and presented in the Spare / Contract Exception section. The shortfall in PO placement is to be broken under subheads like technical evaluation, negotiation with vendor, tender committee, clarifications from vendor etc.

Similarly, shortfall in Indent raising is to be accounted under various initiating department like BMD, TMD, EMD etc.

Section B: OPI – PART A Exceptions

This brings out exceptions in the preparation and submission of Engineering Declaration and its 1st and 2nd revisions. Any other exceptions related to Engineering Declaration and other OPI Part A items needed to be brought to the attention of station management (General Manager, Station) should be brought under this section.

Section C: OPI Scores

Under this section, OPI scores (Part A, Part B – Monthly, Part B – Cumulative) will be brought out. This would reflect how overhaul preparation of the station is progressing. Part A and Part B – Monthly scores should be ideally 100 and any score less than this indicates shortfall in overhaul preparation. It may be noted that, different units will have different OPI no. at a particular time / month. For each unit, there would be three scores namely Part A score, Part B – Monthly score, Part B – Cumulative score. There is a current OPI score column and then there are past scores at various milestones namely OPI 24, OPI 18, OPI 15, OPI 12, OPI 9, OPI 6, OPI 3. Thus OPI scores at various milestones will reflect how overhaul preparation is progressing with time

Section D: Flagged Issues

Issues related to overhaul preparation and that need attention of station GM, ED Region, Operation Services and Director (Operations) should be brought out under this section. For example, a spare turbine rotor stuck up OEM’s Workshop and needing attention / discussion by ED / D(O) for its repair / restoration may be brought out under this section. The details of the specific issues will be brought in the “Remarks” column.

Section E: Key Spares / Contracts for O/H

The detailed status for the key spares and the contracts should be maintained in the format enclosed for planning specific action.

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4.9 OUTAGE COMPLETION REPORT

An outage report will be prepared after the completion of overhauls, compiling all such information, which has relevance to unit overhaul. The report should supplement efforts of overhaul planning and serve as a valuable feedback to future planning process.

Each completion report should bear a unique number which shall be used for identification of unit overhaul as well. The cross reference of the completion report should be entered in the equipment history. The outage completion report shall contain various sections and may be prepared along the following guidelines:

SECTION-1: EXECUTIVE SUMMARY

The section shall contain the constituted outage committee and its members. The section shall highlight the major events (planned vs. actual) and contain the following information:

- Summary of the unit overhauling
- Major work done during overhaul.
- Major spare replacements.
- Major problem faced during overhaul including Constraint and delay.
- Summary of Efficiency improvement after overhaul & Cost Benefit Analysis
- Major works left over during overhaul as per planning due to spare problems or any other reason.
- Pool spares used & action plan details.
- Summary of areas wise contract with scope and values.
- Major achievements during overhaul which directly affects cost of overhaul.
- PERT network & bar chart for all areas e.g. Boiler, Turbine, Electrical, Offsite and C&I. The detailed PERT network (LI, L2, &L3) as planned with actual dates of completion of event shall be incorporated.

SECTION-2: BOILER & AUXILIARIES

This section shall contain all relevant information pertaining to the boiler and its auxiliaries with focus on the following aspects.

- Pressure parts inspection details (Deviation, if any).
- Number of HP joints welded and Radiography of joints.
- Critical checks of equipment and their protocols.
- Major work in rotating equipment / ESP / Ducts and dampers / soot blowers / burners / valves etc.

SECTION-3: TURBINE & AUXILIARIES

This section shall contain all relevant information pertaining to turbine and its auxiliaries with focus on the following aspects.

Brief history of Turbine and their auxiliaries attended during overhaul, critical checks and signed protocols.

SECTION-4: GENERATOR & AUXILIARIES

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This section shall contain all relevant information pertaining to generator and its auxiliaries with focus on the following aspects.

- Work in generator, excitation system and its auxiliaries including signed protocols of the various checks done.
- Brief history of generator including past problems / failures.

SECTION-5: ELECTRICAL EQUIPMENTS

This section shall contain all relevant information pertaining to electrical and its auxiliaries with focus on the following aspects.

- Major equipment overhauled like, transformers / motors / SWGR / Bus duct / Breakers / Relays.
- Repair / replacement / modification, if any.
- Critical checks done of transformers and motors will cover switchyard equipment and associated testing reports.

SECTION-6: CONTROL & INSTRUMENTATION

These sections shall contain the details of jobs taken up on individual equipment and calibration/maintenance Q.A. Sheet of various controls & instruments, protections and interlocks.

- Brief scope of all types of work done in overhaul.
- Brief history of equipment attended during overhaul.
- Calibration details.
- Modification and Replacement of equipment with brief details.
- Reason for modification / replacement.

SECTION-7: OFFSITES AND MISCELLANEOUS

This section shall contain the details of job taken up in offsite, with deviations from planned activities, duly filled in maintenance Q.A. sheets and work progress bar chart (planned vs. actual).

The details of plant betterment jobs, modification, and renovations etc. in the main plant area with suitable drawings, materials, manpower, execution cost as planned and actual shall also form the part of this section.

The details of assistance in the jobs taken from in-house facility e.g. workshop etc. as well as external manufacturing facility used shall also be given.

SECTION-8: RECOMMISSIONING & EFFICIENCY EVALUATION

This section shall cover isolation and recommissioning plan and trial run details of different equipment. The dates of planned vs. actual trial run, reasons for deviations, duly completed recommissioning check sheets etc. It is proposed that recommissioning plan & checklist will be adhered for different equipment during the trial run & commissioning activities. Any deviation from the logical sequence of events or safe

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procedures will be recorded and information transferred in the outage completion report.

After the unit is synchronized and stabilized a performance evaluation test shall be carried out within a period of 15 days. The performance evaluation test and various parameters recorded during pre-outage survey shall be reviewed after unit recommissioning and stabilization and analyzed for improvements.

The efficiency of boiler and turbine heat rate are determined and analyzed for any improvements from P.G. test values.

The section shall have a detailed techno-economic evaluation of overhaul and cost benefit analysis.

SECTION-9: MANPOWER & BUDGETS

This section shall contain details of manpower (planned vs. actual) deployed for each activity in the format. The manpower deployment by station (departmental) as well as by individual contractors shall be given according to the skill. Daily manpower deployment shall be plotted. The details of accidents shall also be included.

The proposed budget for outage shall be compared with actual expenditure on different heads and contractor-wise breakup of the same shall also be furnished. This shall contain the following information.

Area-wise cost of job & Material used.

- Year-wise overhauling cost of 05 years in tabular form (Trend charts)
- Summary in Matrix form for last 05 years for major scope of works, running hours and duration.

SECTION-10: CONCLUSIONS AND RECOMMENDATIONS

The concluding section shall give a detailed account of Lesson learnt from planned activities and their reasons.

Lesson learnt from present overhaul and suggestions, if any for implementation during next overhaul or for other projects to be adopted during overhaul.

The report of individual section/area shall be written by respective area engineer in-charge and submitted to long term planning cell within 15 days of execution of overhaul.

Long term planning shall compile the report update the remaining sections not covered under scope of area engineers and release in the proposed format within a month of completion of overhaul.

5.0 REVIEW

This LMI shall be reviewed by the Head of project, on a 2-yearly basis or any change in the process is identified.

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Annex-I - Detailed Outage Planning Milestones Monitoring Reference Check

Sheet by Outage Manager

Station:	Type Of outage	Outage Number:	Review Date:	Name: Outage Manager:
Unit:		Outage Start Date:	Months for Outage Start:	Name: HoP
24 Months MILESTONES				
#	<i>Milestone, Milestone Description and Purpose, and Check-offs Required to Complete</i>			<i>Responsible Individual</i>
1.	Review Outage Plan, Improvement Project Work scopes (Capital, R&M and O&M) for Consistency With Long-Range Outage and Business Plans and Start Detailed Outage Planning- Preparation of Engineering Declaration R-0.			Outage Manager
Review of the unit long-range outage plans, including but not limited to the 6/ 8-year plan, to validate that intended major scope of outage is aligned with the approved plant Performance Plan, to ensure that support organizations are on schedule to support the outage, and to start detailed planning.				
a.	Ensure Outage Manager has been designated and is assigned to lead outage planning.			HoP
b.	Issue basic outage plan outlining major scope items, budget targets, objectives, planning milestones, assumed critical and near critical paths, and initial outage duration goal.			Outage Manager
c.	Review schedule and status of major work activities, and major material contracts and deliveries against the outage plan and identify any deficiencies (e.g. Raising of Indent as per target Matrix of OPI)			Outage Manager
d.	Review critical equipment and material purchases and deliveries and establish and implement material expediting and shop inspection plans as prudent.			Outage Manager
e.	Review capital and O&M Improvement projects compliance with the Projects Improvement process and identify any deficiencies.			Outage Manager
f.	Review the budget (both total and fiscal year) for adequate funding and identify any deficiencies.			Outage Manager
2.	Issue List of Required Design Change Notices (DCN) /Plant Modification as per OSguidelines.			Outage Manager
A list of all DCNs/Plant modification required for the outage shall be issued for review and comment prior to the outage scope freeze milestone.				
a.	Head of Sections (HoS) submit a anticipated DCN/Plant improvement list (both approved and proposed as per cc-os guidelines) to the Outage Manger.			HoS

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b.	Distribution of list, with required response date, to outage Committee members for further action.	Outage Manger	
3	Establish Scope Identification Team. (outage committee)	HoP	
A standing cross-functional scope identification team will be established to perform ongoing reviews of all sources for anticipated outage scope items. Sources include the work order database, the capital and O&M project listing, lessons learnt, equipment history, RCM recommendation and Knowledge team recommendations. Reviews are to ensure all anticipated scope items are included and approved by management for inclusion in the outage, and items requiring a DCN are processed through the DCN procedure.			
a.	Scope Identification team members are selected.	Outage Manager	
4.	Complete 24 Months - General Items	Outage Manager	
General outage management items.			
a.	Issue outage action item list to correct deficiencies found in any of the above with accountable individuals and due dates identified. <i>(This is the initial issue of the outage action item list which is to be maintained and regularly reviewed until the outage is complete).</i>	Outage Manager	
b.	Issue outage milestone review meeting agenda to include items to be reviewed in each milestone review meeting.	Outage Manager	
c.	Ensure detailed plan and schedule is in place to meet next scheduled milestones, assignments are made to appropriate individuals, and date and time is established for next milestone review meeting.	Outage Manager	

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Unit:			Outage Start Date:		Months for Outage		Name:HoP							
18 Months MILESTONES														
#	<i>Milestone, Milestone Description and Purpose, and Check-offs Required to Complete</i>						<i>Responsible Individual</i>	<i>Date of Completion</i>						
5.	Freeze Outage Capital and O&M Improvement Project Work scope						HoP							
The scope identification team will finalize the listing of capital and O&M improvement projects for approval by the HoP. After approval, the project scope is frozen and under formal scope control - Preparation of Engineering Declaration R-1.														
a.	Capital and O&M improvement project work scope approved by the HoP, added to the outage plan, and placed under scope control.						Outage Manager							
6.	Identify and Freeze Significant Special Scope Items						Outage Manager							
Determine need for significant special scope that requires unique skills, material, planning, or other resources that have high potential to impact outage performance. These should also be added to the Risk and Contingency Action Planning Report with potential impacts and mitigating action plans identified. Examples include but are not limited to														
• Acid cleaning of boilers • De-watering for repair, access, or inspection (<i>consider condition of gates, guides, and seals</i>) • De-watering of cooling water tunnels for repair, access, or inspection • Other unique or special testing, inspection, or repair requirements														
a.	Obtain HoP approval of special scope items and incorporate into major work scope listing.						Outage Manager							
7.	Freeze DCNs/Plant modification and Release Any Remaining DCN s for approval.						HoP							
Freeze DCNs/Plant modification and release remaining DCN/ Plant modification packages for approval, if any). DCNs/ Plant modification proposed after this date must be submitted using an Outage Scope Change Request and approved by the Outage Manager.														
a.	Agreement from all groups that DCN/ Plant modification list has been reviewed and is complete, the scope of each DCN/ Plant modification is clearly defined, and that all packages have been identified.						Outage Manager							
b.	Issue of DCN/ Plant modification list by noting that list is now frozen.						Outage Manager							
c.	Verify that formal notification is given to appropriate group(s) to proceed with the work necessary to implement the DCN/ Plant modification .						Outage Manager							
8.	Issue Remaining Long-Lead Engineered Equipment and Material Purchase Requisition (PR)						Outage Manager							

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Each additional work scope item will be reviewed to identify, and list needed material purchases and to help ensure that PRs are issued on time. The Area Maintenance Engineer (AME) e.g., member of the outage committee will inform the Outage Manager of any known problems with missing PRs or delivery dates. The material needs listing will be regularly updated and reviewed in pre-outage planning meetings.

a.	List of long-lead material for each new work package.	AME	
b.	Issue of material list including PR number and issue date and requested delivery dates.	AME	

c.	Addition of agenda item to regular outage planning meeting for review of material status.	Outage Manager	
9.	Identify Needs and Implement Plan to Obtain Any Remaining Needed Information for Vendors, Engineering, or Implementers	Outage Manager	

Review work scope to identify needed as built or special rigging information required for engineering, construction, or vendors and adds actions to collect needed information to outage action item list. Create shutdown work orders to ensure that opportunities to gather information when the unit is offline are not missed.

a.	Add specific actions necessary to obtain needed information to the outage action item list with accountable individuals and due dates identified. <i>(consider use of shutdown WOs where appropriate)</i>	Outage Manager	
10	Complete Months 18 - General Items	Outage Manager	

General outage management items.

a.	Complete 18 Months - status review.	Outage Manager	
b.	Review outage action item list and add any identified deficiencies.	Outage Manager	
c.	Ensure detailed plan and schedule is in place to meet next scheduled milestones, assignments are made to appropriate individuals, and date and time is established for next milestone review meeting.	Outage Manager	

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lant:		Type OF outage	Outage Number:		Review Date:		Name: Outage Manager:	
Unit:			Outage Start Date:		Months for Outage		Name:HoP	
12 Months MILESTONES								
#	<i>Milestone, Milestone Description and Purpose, and Check-offs Required to Complete</i>						<i>Responsible Individual</i>	<i>Date of Completion</i>
11	Update Generic Schedule and Incorporate Pre-outage Planning and Key Project Milestones						Outage Manager	
Outage Manager will ensure that the baseline schedule is updated to reflect latest procedure changes, lessons learned from similar outages, and is expanded to incorporate pre-outage planning and project activities and milestones. The updated schedule becomes the Engineering Declaration R-2.								
a.	Confirm that the Generic schedule has been updated to reflect latest shutdown and logic, and recurring routine outage preventative maintenance (PM) tasks.						Outage Manager	
b.	Perform comparison of other POWER UTILITY units for lessons learned in shutdown, startup, and recurring PMs.						Outage Manager	
c.	Incorporate pre-outage and project activities and milestones.						Outage Manager	
12	Complete Months 12 - General Items						Outage Manager	
General outage management items.								
a.	Review outage action item list and add any identified deficiencies.						Outage Manager	
b.	Ensure detailed plan and schedule is in place to meet next scheduled milestones, assignments are made to appropriate individuals, and date and time is established for next milestone review meeting.						Outage Manager	
#	<i>Milestone, Milestone Description and Purpose, and Check-offs Required to Complete</i>						<i>Responsible Individual</i>	<i>Date of Completion</i>
13	Establish Pre-Outage Organization and Begin Monthly Pre-Outage Planning Meetings						HoP	
HoP ensures each POWER UTILITY partner, and contractor organization with a significant outage role identifies primary and alternate representatives. Representatives will have decision authority for outage issues related to their organization or project and will be responsible for their portion of individual milestones. Outage Manager initiates monthly pre-outage meetings to address readiness status, problem areas and threats, thoroughness of options considered for cost, schedule, and quality improvements and corrective action plans. Widely dispersed organizations occasionally hold meetings via telecom. HoP shall attend the Meeting. Meetings have written agenda which builds on the previously developed milestone meeting agenda and includes detailed review of unresolved outage issues and the outage action item list.								
a.	Identification of the organizational representatives.						HoP.	

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b.	Development of monthly meeting agenda, meeting calendar, and completion of first meeting.	Outage Manager	
14	Develop Outage Task and Service Contract Plan	Outage Manager	
.	The Outage Manager ensures that a contracting plan is in place which lists work packages to be contracted. Plan contains scheduled purchase request release and award dates. After plan issue these dates shall be added to the pre-outage schedule. Outage work area Engineer submits the detail of necessary PRs. They also ensure requested contracts are budgeted, contractor schedules and plans are incorporated into overall outage planning, and contractors understand pre-outage and outage schedule commitments.		
a.	Contractor plan developed and issued.	Outage Manager	
b.	PR release and contract award/ required dates added to overhaul schedule.	Outage Manager	
15	Issue DCNs/Plant modification	Outage Manager	
.	Verify issue of all DCNs and receipt of documentation required to make changes and to conduct necessary training, calibration, and testing.		
a.	All documentation issued and received.	Outage Manager	

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Unit:		Outage Start Date:		Months for Outage		Name:HoP	
09 Monthly MILESTONES							
#	<i>Milestone, Milestone Description and Purpose, and Check-offs Required to Complete</i>					<i>Responsible Individual</i>	<i>Date of Completion</i>
16	Issue overhauling Schedule					Outage Manager	
This milestone reflects the first formal issue of the outage and pre-outage schedule. It will be based on the generic schedule which includes the work and testing performed in every outage. This schedule includes preventive and corrective maintenance, major capital and O&M work and any other work approved to date. Also planning activities tied to pre-outage milestones and significant capital and O&M project milestones shall be included. This revision will provide the framework for further outage planning & schedule and management control of pre-outage activities.							
a.	Issue and distribute the overhaul schedule to all outage participants for use and review.					Outage Manager	
17	Incorporate Lessons Learned from Previous Outages into Outage Planning					Outage Manager	
Post-outage critique actions from previous outages and identified corrective actions and enhancements should be incorporated to improve outage schedules and performance. A composite listing of applicable lessons learned/improvement actions will be							
a.	Completion of review of lessons learned database and incorporation of actions necessary to implement into outage action item list.					Outage Manager	
18	Issue Bills of Material For Long-Lead Non-Engineered Material and Reserve All Identified Critical Parts					Outage Manager	
Each outage work package E IC shall reserve critical parts and identify requirements for long-lead non-engineered material purchases (delivery greater than 6 Months from the control of purchase requisition). Procurement will be informed of any items that are missing and items that are needed but have unacceptable delivery dates. Procurement will begin reporting at regular outage planning meetings any material delivery date problems. This milestone starts the verification of materials on hand. Progress shall be monitored and reviewed as an agenda item in the regular outage planning meeting.							
a.	Issue to performing organizations of sufficient design detail to develop PRs for long- lead non-engineered material.					EIC	
b.	Acknowledgement by Outage Work Package EIC to procurement that the required material needs have been identified and the delivery dates are acceptable					EIC	
c.	Acknowledgement by Outage Work Package EICs that identified critical parts have					EIC	
d.	Addition of long-lead non-engineered material requirements to the material needs list.					Outage Manager	

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19	Complete Initial Work Package Development for All Approved Outage Scope and Issue Requests for Proposals/Estimates.	Outage Manager	
This milestone marks the issue of requests for proposals/estimates from potential implementing organizations. Initial work packages covering all approved scope have been developed and contain sufficient detail and design information to allow proposals/estimates to be completed to support work authorizations. Exceptions, with recovery plans, will be noted and approved by Outage manager and added to the outage action item list.			
a.	Issue to performing organizations of sufficient design detail to develop authorization quality estimates.	Engg.	
b.	Formal issue of requests to potential implementing organizations for all approved outage scope.	Outage Manager	
20	Complete Months 09 - General Items	Outage Manager	
General outage management items.			
a.	Complete Months 09 - status review.	Outage Manager	
b.	Review outage action item list and add any identified deficiencies.	Outage Manager	
c.	Ensure detailed plan and schedule is in place to meet next scheduled milestones, assignments are made to appropriate individuals, and date and time is established for next	Outage Manager	
21	Freeze Corrective and Preventative Maintenance (PM) Work Orders (WO) and Inspections	HoP	
This milestone freezes routine maintenance work and inspections. Scope identification team shall use the latest version of the PM and WO backlog database, work identified in system walk downs and from input from knowledgeable individuals to perform line item evaluations of proposed outage work. Reviews consider if the work must be done during the outage. Work proposed after this date must be submitted using an Outage Scope Change Request.			

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Unit:		Outage Start Date:	Months for Outage	Name:HoP	
06 Monthly MILESTONES					
#	<i>Milestone, Milestone Description and Purpose, and Check-offs Required to Complete</i>			<i>Responsible Individual</i>	<i>Date Of Completion</i>
23	Conduct Review and Approve Outage Scope			HoP	
Milestone completion includes review and approval of outage scope by HoP for inclusion in the overhauling schedule. Scope is based on previously frozen major and routine maintenance work scope and approved scope changes. All work groups have verified need for maintenance and calibration activities and have written needed work requests. Work requests for contingency planning and temporary support (scaffolding, insulation, power, and temporary and special equipment) have also been written.					
a.	Confirmation by the HoP that the above actions have been completed and the scope and budget is approved for inclusion in the schedule.			HoP	
24	Issue Overhauling Schedule and Begin Fortnightly Outage Planning Meetings.			Outage Manager	
Milestone reflects conclusion of review of the Overhauling schedule and associated logic and incorporation of comments and will include detailed activities for all known pre-outage and outage scope and contain outage critical capital and O&M project milestones.					
a.	Confirmation that comments on OH Schedule including logic changes have been submitted to and resolved with Outage Management and the organization responsible for the work activity			Outage manager	
b.	Confirmation that the schedule contains all approved outage work and detailed pre- outage activities.			Outage Manager	
c.	Issuance of OH schedule for review.			Outage Manager	
d.	Development of weekly outage meeting agenda and calendar, and completion of first weekly meeting.			Outage manager	
25	Identify Budget Issues Requiring Resolution			Outage manager	
Continue the process of balancing the best available outage forecast based on current proposed scope with the approved budget. Identify issues that must be addressed to finalize scope and budget and assign action items to resolve.					
a.	Identification and inclusion of all actions required to finalize budget into outage action item list.			Outage Manager	

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26 .	Verify Availability of Identified Critical Reserved Parts	Outage manager	
This milestone starts the verification of parts on hand to meet outage commitments.			
a.	Outage manager verification that the parts on hand verification process has started.	Outage Manager	
27 .	Complete Months 06 - General Items	Outage Manager	
General outage management items.			
a.	Complete months 06 - status review.	Outage manager	
b.	Review outage action item list and add any identified deficiencies.	Outage manager	
c.	Ensure detailed plan and schedule is in place to meet next scheduled milestones, assignments are made to appropriate individuals, and date and time is established for next milestone review meeting.	Outage Manager	

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Unit:			Outage Start Date:		Months for Outage start:		Name: HoP	
03 Months MILESTONES								
#	Milestone, Milestone Description and Purpose, and Check-offs Required to Complete						<i>Responsible Individual</i>	<i>Date of Completion</i>
28.	Complete Schedule Optimization Reviews with Implementing Organizations						Outage manager	
Completion of detailed schedule reviews with implementing organizations to ensure that least duration options and associated costs and benefits have been identified for critical and near critical path activities and to identify weaknesses in pre-outage/outage activities that need attention. Review results will be used to support executive approval of duration target and will be reflected in OH schedule.								
a.	Reviews complete and benefit and cost analysis of outage duration options developed.						Outage manager	
b.	Identified deficiencies added to outage action item list.						Outage manager	
29.	Complete Outage Mobilization, Lay down, and Heavy Lift Plans						Outage manager	
Site-wide plans shall be established for mobilization of personnel, lay down of equipment and material, and conduct of heavy lifts. Implementers shall provide conservative estimates of training and in-processing needs. Outage team must ensure that processes and facilities can support the expected influx of personnel.								
a.	Identification of number of people that must be processed and processing requirements.						AME	
b.	Completion of development of mobilization plan including provisions for restrooms, eating areas, lay down and fabrication areas, security, ingress and egress routes, etc.						Outage manager	
c.	Completion of development of heavy lift plans including travel routes.						Outage manager	
30.	Assign Operations Outage Clearance and Return to Service Team (Commissioning Team)						HoP	
Operations will assign a lead person to begin clearance and return to service planning for the outage. This will include review and identification of scope for the clearance and clearance boundaries, and the activities and logic necessary to return the unit to service especially considering any DCN return to operability (RTO) / recommissioning tests or special requirements. This milestone activates the team and starts the planning for needed clearances and return to service activities which will be used as input to the OH schedule.								
a.	Assignment of operations outage clearance and return to service team lead with schedule of activities to be accomplished including any DCN / RTO requirements.						HoP	

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31. Develop Material Delivery Exception List	Outage manager	
Procurement will report delivery problems as soon as known. Schedule impacts are identified by outage work package EIC for delivery dates beyond the need date for pre-outage work or one month prior to outage start for outage work. All participants must inform Procurement of any material that has been added, deleted, or rescheduled. Periodic report updates will be provided.		

a.	Develop and distribute material exception list.	AME	
b.	Add review of material exception list to outage meeting agenda.	Outage Manager	
32. Identify Craft Labor Needs		Outage manager	
Craft labor requirements required to accomplish outage and pre-outage workshop must be communicated to and rolled up by the Outage EIC. The Outage EIC ensures latest information is communicated for final system-wide labor resource availability analysis.			
a.	Identify individual work package labor needs.	Outage manager	
33. Obtain HoP Approval of Outage Scope, Budget, and Duration		HoP	
Final approval of outage scope, schedule, and budget following a rigorous review by HOD-O&M. Review verifies that estimates are appropriate, least schedule duration options are identified, comprehensive risk analysis completed, work scopes fix targeted problems, and scope which could be worked non-outage is minimized. After consideration of the costs, benefits, and risks the target outage duration is set.			
a.	Obtain final approval of outage scope and budget and duration targets from HoP.	Outage Manager	
34. Issue Outage Organization Structure Including Executing Departments / Agency		Outage Manager	
Identify by name and function the key organizations and organizational persons responsible for all facets of implementation and including plant departments, Engineering, Construction, Workshop, Partners, contractors, and key support personnel.			
a.	Issue of the outage organization chart including functional accountabilities.	Outage manager	
35. Complete Months 03 - General Items		Outage manager	
General outage management items.			
a.	Complete month 3 - status review.	Outage Manager	
b.	Review outage action item list and identified deficiencies till resolved.	Outage manger	
c.	Complete Standard Pre shutdown check list of executing department and agencies	AME	

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Annex-III - ENGINEERING DECLARATION FORMAT (Typical)

_____ Super Thermal Power Station

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- Unit No.:
- Boiler License No:
- Boiler License validity (date):
- Overhaul History (In reverse chronological orders) for the past five years:

S. No.	Description	Overhaul duration			Overhaul Category	Operating Hrs. since prev. OH	Cumulative operating Hours	Forced Outages Between two overhauls (%)			Important & major high lights
		From	To	Total				BTL	Trip	Total	
1	Proposed overhaul		---	-----							
2	Previous overhauls										
3	----- ---										
4	----- ---										

5. Unit Milestones /Major Event since inception:

S. No.	Milestones/Events	Data	Remarks
1.	Firs synchronization (date)		
2.	First commercial (date)		
3.	Longest operating period (date /duration in hrs.).		
4.	Min. BTL (Nos.) achieved between two overhauls		
5.	Min Trips (Nos.) achieved between two overhauls		
6.	Min forced outage (%) achieved between two overhauls		

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6. Unit performance History of Past five years.

S. No.	Parameters	Current year till date	Previous four years			
1	Generation (MU)					
2	Availability (%)					
3	Plant load factor (%)					
4	Loading factor (%)					
5	Partial loading (%)					
6	Aux. Power Consumption					
7	(%) Make up (%)					
8	Heat rate					
9	No. of BTL					
10	No. of Trips.					
11	Total forced outage (%)					
12	BTL as % of F.O.					
13	Trip as % of F.O.					

7. Root Cause Analysis on BTL & Action Plan – Past Five years

S. No.	Recommendation	Date of recommendation	Implementation status	Effect of Action or Improvement achieved
1.	Action plan for present overhaul			
2.	Actions of previous overhauls (Past five years)			

8. Root Cause Analysis on Trip & Action Plan-Past Five years.

S. No.	Recommendation	Date of recommendation	Implementation status	Effect of Action or Improvement achieved
1.	Action plan for present overhaul			
2.	Actions of previous overhauls (Past five years)			

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8a. Top 10 equipment based on Risk Plot prepared with 5 yrs. history of forced outage & partial loading:

Top 10	Name of the equipment	Risk coverage %	Action plan Routine	Action plan Additional	Included in O/H scope at point no.- ---	Remarks if any
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

8b. Top 10 equipment based on Risk Grid as updated on -----mention month /yr. of updation):

Top 10	Name of the equipment	Risk coverage %	Action plan	Included in O/H scope at point no.- ---	Remarks if any
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

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09. Major Equipment failure History (between two overhauls) for past Five years

S. No.	Overhaul Duration	Equipment	Failure/analysis/corrective action plan/ATR etc.
1.	Current year		
2.	Previous Years		
3.			
4.			
5			

10. Unit/Equipment Performance & Efficiency
Evaluation and Cost Benefit Analysis Parameters

S. No.	Description /Parameter	Values				Expected. Improvements	
		Norms	Present	Variance	Expected after OH.	Tech. Gains	Financial Gains
1	MS emp/pressure						
2	Heat Rate (Kcal/						
3	kwh) APC (%)						
4	Make up (%)						
5	Turbine efficiency.						
	• HPT						
	• IPT						
	• LPT						
6	Boiler Efficiency						
7	HPH FW (in/out)						
8	APH Flue gas (in/out)						
9	O ₂ in flue gas (%)						
10	Condenser vacuum (mm of Hg)						
11	CW Temperature (in/out) (OC)						
12	Mills fineness (%)						
13	Mills output. (T/Hr)						
	Mill Motor current (amps)						
14	Equipment performance e.g. Motor current & discharge pressure etc.						
	• BFP						
	• CEP						
	• ID Fan						
	• FD Fan						
	• PA Fan						

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11. Work Module:

Enclose list of equipment along with jobs to be taken up under the following categories:

- a. Standard work modules
- b. Plant betterment plans (Modifications & R&M)
- c. Pending defect & PM jobs
- d. Corrective / preventive action plans arising out of trip analysis, BTL analysis, Failure diagnostics, Reliability Centered Maintenance (RCM), Risk Evaluation and other technical audits points.
- e. Action Plan for net-efficiency(Heat rate & APC) improvement

Sr No .	Equipment with efficiency gap (heat rate & APC)	Action Plan Routine	Action Plan Additional	Included in O/H scope at point no.- ---	Remarks if any

Justification for taking up various jobs (e.g. operational inadequacies, statutory requirement, Major plant modifications etc) in quantified terms and its impact on the equipment performance and reliability has to be furnished.

This could be along the following lines:

1.0 Operational inadequacies:

AI=Availability Improvement RI = Reliability Improvement EI = Efficiency Improvement

2.0 Statutory requirements:

SR= Statutory Requirement EN=Environmental inadequacies

3.0 Major plant modifications:

TO=Technological Obsolescence

LF= Replacement of major plant components due to life extension plans

12. Comments/Confirmation regarding recommendations of Knowledge Teams/ Corp OS/ RHQ-OS being included in the Scope of Work:

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13. Financial Justification/Evaluation of Overhauls

Overhaul Year	Budgeted/Actual Expenses			Payback period (Months)	Gains (*)	
	Services	Material	Total		Direct	Indirect
Proposed Previous Five Years						

14. Comments/Confirmation regarding Identification & Procurement/Award Plan of Spares/Cons/T&Ps/Contracts:

15. Enclose the followings: -

- PERT network & bar chart
- Exception report of condition monitoring on all major equipment undergoing overhaul where vibrations & other parameters indicate abnormal condition and need of intervention.

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Annex – IV - Checklist for ORT Coordinator

01. Availability of O/H Plan for the current financial year (Annual Plan) and 6-Year/8-Year Rolling Plan for O/H
02. Confirmation regarding the date of Preparation/Submission of Engg Declaration and its Revision by the site for the O/H scheduled for current financial year, next financial year and next to next financial year as per the annual plan / 6/8-year rolling plan.
03. Confirmation regarding Outage In charge and Outage Committee formation
04. Check/discuss with site if the Target Matrix is realistic & feasible and bring out exception during ORT for needed measures/correction.
05. Review of OPI Score Calculation Format Part-A and Part-B filled up by Site Outage Committee for O/H of each unit scheduled at the station for current FY, next FY and next to next FY(-24 months OH planned).
06. OPI Summary Sheet for ORT to be reviewed by ORT Co-Ordinator in the given format for O/H of all units scheduled at the station for current FY, nextFY and next to next FY.
07. List of Critical Spares Area-wise, Contracts and Miscellaneous Issues pertaining to the O/H of each unit scheduled at the station for current FY, next FY and next to next FY(-24 months of unit OH planned) and their status. This is to be prepared by outage manager/MTP/ Outage Committee and reviewed by the ORT.
08. ORT coordinator would generate Exception Report in prescribed format for Intervention of Region required to ensure readiness for O/H as per schedule for current FY, next FY and next to next FY.

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Annex-V - Monthly Exception Report (sample Sheet)

Overhaul Preparedness Index (OPI)								
Station		Monthly Exception report						
Overhaul: for FY		Sample Sheet						
A)		Resource (Material & Contract) Exceptions						
S.No.	Activity Pending wrt Target Matrix	Pending for current OPI		Pending since previous OPI Review (more than 1 Month of Target)		Pending before previous OPI Review (more than 2 Months of Target)		Remarks
		Materials	Contracts	Material	Contracts	Materials	Contracts	
	Indent raising	13						
	BMD	5	1	3		1		
	TMD	2		2		0		
	EMD							
	C&I	5	1	3		2		
	Offsite Mtnc	1						
	LTP - MTP							
	Issue of enquiry pending	2		2				
	C&M							
	O&M							
	PO placement	22	1	12		8		
	Non receipt of bid	2						
	Technical Evaluation	10	1	6		4		
	Negotiations	2		1		1		
	TC	3		3				
	Vendor	4		2		2		
	Retendering	1		0		1		
	Receipt of material	8		2		6		
B)	OPI - PARTA Exceptions	No. of pending items		Pending since previous OPI Review (more than 1 Month of Target)		Pending before previous OPI Review (more than 2 Months of Target)		Remarks
	Engg. Declaration Submission							
	R0							
	R1							
	R2							
	Others:							
Overhaul Preparedness Index (OPI)								
Station		Monthly Exception report						
Overhaul: for FY		Sample Sheet						

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C)	OPI Scores						Meeting Date:				
Unit no.	OPI no.	Score	Current OPI score %	Scores at diff. milestones %							Remarks
				OPI 24	OPI 18	OPI 15	OPI 12	OPI 9	OPI 6	OPI 3	
1	-1	Part A	100	95	90	90	85	80	80	89	
		Part B monthly	95	80	70	70	75	80	80	85	
		Part B cumulative	95	2	15	15	35	55	55	85	
2	-5	Part A	95	87	88	90	85	90	95		
		Part B monthly	90	85	80	90	83	95	90		
		Part B cumulative	75	10	20	30	50	63	70		
3		Part A									
		Part B monthly									
		Part B cumulative									
4		Part A									
		Part B monthly									
		Part B cumulative									
5		Part A									
		Part B monthly									
		Part B cumulative									
6		Part A									
		Part B monthly									
		Part B cumulative									

D)	Flagged Issues	Issue details	Description / Remarks
	GM Station		
	ED Region		
	ED OS		
	Director Operations		

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Annexure-VI - Suggestive Pre-Overhauling Checklist

SUGGESTIVE PRE-OVERHAUL CHECK LIST			
SN.	Activity	Action By	Status
1	Servicing of EOT Cranes, Hoists (including APH hoist) & Rigging Equipment	RM /PP/EMD / TMD	
2	Healthiness checking of EOT crane limit switches	EMD/SAFETY/TMD	
3	Testing of Agencies welding M/Cs	All Mant./EMD/SAFETY	
4	Testing of Agencies electrical portable appliances	EIC/EMD/SAFETY	
5	Agencies tools & tackles and safety harness healthiness	EIC/SAFETY	
6	Healthiness of extension boards fitted with ELCBs	EIC/EMD	
7	Meeting with contractors & Agency safety supervisors in presence of EIC & Head of O&M	EIC/SAFETY	
8	Safety inspection / survey & submission of report	SAFETY	
9	Contractors tools load test to ensure use of Tested Tools by Contractors	All Mtc & Safety	
10	Arrangement of drinking water at all location /resting space for work man	Civil / All Mtc.	
11	Ensuring availability of dark room for developing of RT films.	PP	
12	Availability of one complete set of scaffolding for main boiler scaffolding and ensuring its healthiness.	PP	
13	Availability PVC/Metallic Caps for boiler tube	PP	
14	Servicing / Checking of all Ladders & Scaffoldings prior to use for Boiler Survey	PP	
15	Burners / Coal Pipe shifting	PP	
16	Scaffolding Shifting. Servicing of Scaffolding Clamps	PP	
17	Identification & Marking of Defective Steam Traps for replacement during SD	PP/ OPN	
18	APH Baskets high pressure water jetting Point to be identified	RM / OPN	
19	To ensure availability of Argon Gas. All Argon Cylinders to be tested at site for purity	PP / TMD /Chem	
20	Heater required for Electrodes pre heating	PP / TMD	
21	Inspection of Trench & its Covers in Boiler area	Civil	
22	Ready cut shims		
23	O2 & DA Gas availability with contractors	All maint. deptt	
24	Approach Platforms & structures material availability	All Mtc/ Opn	
25	Checking of existing Welding Points & ensuring healthiness.	EMD	
26	Healthiness of Boiler Lift, load testing & phone availability ensuring availability & manning round the clock	EMD/IT/MTP/OPN	
27	TG Hall Roof Light Checking	EMD/OPN	
28	Availability of Control & Power Fuses is to be ensured	EMD/OPN	
29	Preparation of Gate Passes	HR/All Mtc.	

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30	Telephone / LAN Connections in Boiler & Turbine Area	Respective area + IT	
31	Empty drums to be kept readily available before O/H	RM,TG, AHP	
32	Trip committee/Energy Audit recommendation	EEMG	
33	Efficiency Testing Before O/H.	EEMG	
34	Formation of Quality Team & Materials movement Teams for Unit overhaul	MTP	
35	Note Sheet Initiation for Approval of Tea & Snacks for OH	MTP	
36	Work Plan Activity Bar Chart Preparation for OH	MTP & All Mtc	
37	Pre-Outage Vibration Survey & Recommendation	MTP-CM	
38	Zero forced outage action plan	MTP & All Mtc, EEMG	
39	Knowledge Team Recomm action plan	MTP & All Mtc, EEMG	
40	Technical Audit pt. action plan	MTP & All Mtc, EEMG	
41	Formation of Commissioning Team for Unit overhaul	AGM / DGM(O)	
42	Dewatering arrangement during Boiler water washing	Operation	
43	Arrangements for External washing of Boiler	Operation	
44	Placement of Extra Lockout Boxes in UCB	Operation	
45	Actual Over speed Testing before synchro. the Turbine is to be done	Operation	
46	Soot blowers operation	Operation	
47	Emptying of Bunker/feeders before stopping the unit	Operation	
48	Pre Outage Survey – Steam/Air Leakages & all defects to be identified	OPN / All Mtc.	
49	List of valves passing / leakages & Valves to be taken up for OH	Opn, PP, TMD, AHP	
50	Hard copy of blank protocols, check lists, commissioning check lists customized for the unit , to be submitted to MTP before overhaul.	OPN	
51	Isolation & PTW Planning & Ensuring issue of All PTWs through SAP	OPN /MTP / All Mtc./Safety	
52	Polythene Covers for C&I Panels & instruments & Elect. Motors & JB's for Boiler External Water Washing	C&I & EMD	
53	Accommodation to contractors in GH/FH/Quarters	MTP / All Mtc /HR	
54	Alternate supply to Boiler lifts	EMD	
55	Arrangement of vehicle for mobilization Man power/material	HR/All Mtc.	
56	Formation of team for scrap removal from working site.	Respective Deptt.	
58	Quality of Argon gas in terms of its purity 99.9996% to be ensured to eliminate porosity problem in the weld.	BMD	
59	Hydro ejector arrangement to be done prior to OH for penthouse evacuation	BMD	
60	Arrangement for RH pressurization test	BMD	
61	Arrangement of air grid across Boiler	BMD	

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62	Arrangement of 24volt DC in the Boiler area.	BMD	
63	Readiness for the arrangement of sand blasting	TMD	
64	Readiness for turbine rotors stand	TMD	
65	Readiness of arrangement for chemical cleaning	BMD	
66	Readiness of arrangement for APH basket placement & cleaning	BMD	
67	Readiness of Mechanical/ Electrical workshop including availability of workshop equipment's to be made ready before of start of unit Overhauling	BMD/TMD/EMD/AHD	

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Annexure- VII -Suggestive List of Overhauling Spares-Boiler

A. Boiler (500 MW)

Pressure Parts (500MW)

S.No.	Location	DESCRIPTION	Min Req'd Qty	Remarks
1	Economizer	Economizer straight tube of each type, size, thickness & material	300 mtrs	May change depending on past erosion rate
		Economizer Hanger Tubes of all sizes, thickness & material	100 mtrs.	
		Economizer Tube Bends - 180 Deg. Bends	30 Nos.	
		Economizer Tube Bends - 90 Deg. Bends of various type and sizes	20 Nos.	
		Economizer inlet header Tube Bends	20 Nos.	
		Economizer hanger tube bends	20 Nos.	
		Economizer hanger swage tubes	20 Nos.	
		Cassette Baffles of various sizes	20 Nos.	
		Shields of various sizes	1000 M	
		90 Deg. Bend Shields	60 Nos.	
		Chicken wire mesh	300 mtrs	
		Saddle straps	50 Nos.	
2	Water Walls	Water wall straight tube of each type size thickness and material (RIFLE WATER WALL TUBES)	300 mtrs.	May change depending on erosion rate
		Water wall plain tube of each type size thickness and material for water wall screen, water wall hanger tubes, waterwall extended wall etc.	125 mtrs.	
		Water Wall tube bends of various types, sizes for openings such as Scaffolding doors, Peepholes, Manhole Doors, LRSBs,	3 Nos.	
		Water Wall tube bends of various types, sizes for openings such as Wall Blowers	40 Nos.	
		Water Wall tube bends of various types, sizes of Goose neck	20 Nos.	
		Water Wall tube bends of various types, sizes for S Panel	30 Nos.	
		Water Wall pull out bends	3 Nos.	
		Water wall tube bends of each types, sizes, material for Burner area	3 Nos of each type	
		Swage tube for rear water wall	25 Nos.	
		Steam cooled wall of each type size thickness and material	100 mtr.	
		Backpass screen tubes of each type size thickness and material	150 mtrs..	

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	Steam cooled wall bends of each type size thickness and material - Manhole tube bends, LRSB opening bends, bifurcating bends (Fork tubes), Bends near C headers	10 Nos. of each type	
	Straight Shields of various sizes for back pass screen tube	200 mtrs	
	Shields Bends of various types, sizes for back pass screen tube	20 Nos.	
	LTSH straight tube of each type size thickness and material	100 mtrs.	
	LTSH tube bends of each type size thickness and material - 90 Deg., 180 Deg, 50 Deg., Offset bends, Double Offset bend, S Bends, Bifurcating bends	10 Nos.	
	LTSH shield bends of each type, sizes	10 Nos.	
	Saddle straps	100 Nos.	
	LTSH Hanger tubes	100 mtrs.	
	LTSH straight shields of each type, size	200 mtr.	
	Div. panel Super Heater straight tube of each type, size, thickness and material	100 mtrs.	
	Div. panel Super Heater tube bends of each type size thickness and material - 90 Deg., 180 Deg.	10 Nos.	
	Connectors and Lugs	50 Nos.	
	Platen Super Heater Straight Tube of each type size thickness and material	100mtrs	
	Spacer tubes/ Binder tubes	50 Mtr.	
	Spacer tube bends	2 Nos.	
	Radiant roof tubes	100mtrs	
	Each type of Liner for Desuperheater of Superheater	1 No.	
	Platen Super Heater tube bends of each type size thickness and material - 90 Deg., 180 Deg.	10 Nos.	
	RH straight tube of each type, size, thickness and material	150 mtrs..	
	RH tube bends of each type, size, thickness and material	10 Nos.	
	Dissimilar metal weld joint spot piece for SH and RH of each type size thickness and material	10 Nos.	
	Manhole gasket for Drum and for bottom ring headers	10sets	
	Orifice for bottom Ring header	1 Set	

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	Fasteners of drum such as J bolts etc.	1 Set	
	Complete pump and motor assy	1 No.	
	Journal bearing	1 set	
	Thrust bearing	1 set	
	Casing wear ring	1 set	
	Impeller wear ring	1 set	
	Coal compartment assembly	2 sets	
	F.I.Elbow	10 Nos.	
	Victauc Coupling for fuel pipes	10 Nos.	
	Fuel oil guns	4 nos	
	Hoses for Oil guns	40 Nos.	
	Fuel oil burner isolation valves	4 nos	
	Gaskets for various sizes for Burners	1 Set	
	Ceramic Insulations pads for Burners		
	Fasteners of various sizes for Burners	1 Set	
	Complete boiler drum safety valve of each type	1 no.	
	Upper adjusting ring for boiler drum safety valve of each type	1 nos.	
	Lower adjusting ring for boiler drum safety valve of each type	1 nos.	
	Safety valve disc for boiler drum safety valve if each type	2 nos.	
	Safety valve Spindle for boiler drum safety valve of each type	1 no.	
	Safety valve guide for boiler drum safety valve of each type	1 nos.	
	Safety valve locking pin set for boiler drum safety valve of each type	1 set	
	Complete SH safety valve of each type	1 no.	
	Upper adjusting ring for SH safety valve of each type	1 nos.	

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	Lower adjusting ring for SH safety valve of each type	1 nos.	
	Safety valve disc for SH safety valve of each type	2 nos.	
	Safety valve Spindle for SH safety valve of each type	1 no.	
	Safety valve guide for SH safety valve of each type	1 nos.	
	Safety valve locking pin set for SH safety valve of each type	1 set	
	Complete RH safety valve of each type	1 no.	
	Upper adjusting ring for RH safety valve of each type	1 no.	
	Lower adjusting ring for RH safety valve of each type	1 no.	
	Safety valve disc for RH safety valve of each type	2 nos.	
	Safety valve stem for RH safety valve of each type	1 no.	
	Safety valve guide for RH safety valve of each type	1 nos.	
	Safety valve locking pin set for RH safety valve of each type	1 set	
	Complete ERV valve assembly including pilot valve for SH of each type	1 no.	
	Disc for ERV main valve for SH of each type	2 nos.	
	Spring for ERV main valve for SH of each type	2 nos.	
	Seal rings for ERV main valve for SH of each type	2 nos.	
	Seat bushing for ERV main valve for SH of each type	2 nos.	
	Disc and stem assy for ERV pilot valve for SH of each type	2 nos.	
	Bushing for ERV pilot valve for SH of each type	2 nos.	
	Spring for ERV pilot valve for SH of each type	2 nos.	
	Complete ERV including pilot valve for RH of each type	1 no.	
	Disc for ERV main valve for RH of each type	2 nos.	
	Spring for ERV main valve for RH of each type	2 nos	
	Seal rings for ERV main valve for RH of each type	2 nos	

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	Disc and stem assy for EV pilot valve for RH of each type	2 nos	
	Bushing for ERV pilot valve for RH of each type	2 nos	
	Spring for ERV pilot valve for RH of each type	2 nos	
	Seal ring for ERV pilot valve for RH of each type	2 nos	
	Pressure seal gasket for Main steam stop valve	2 nos	
	Stem for Main steam stop valve and its integral bypass	2 nos	
	Disc for Main steam stop valve and its integral bypass	1 no	
	Seat ring for Main steam stop valve and its integral bypass	2 set	
	Complete SH spray control valve assy	2 No.	
	Complete RH spray control valve assy	2 No.	
	Class 1500 gate/globe valve assy of sizes 25mm/ 40mm/ 50mm	10 nos	
	Class 2500 gate/globe valve assy 25mm/ 40mm/50mm	10 nos	
	Class 2500 gate/globe valve assy 65mm/80mm/100mm	5 Nos.	
	Complete assembly of water wall deslagger	4 nos	
	Complete valve assembly for water wall de- slag	6 nos	
	Bearings for Soot blowers	1 Set	
	Complete assembly of LRSB	2 nos	
	Complete valve assembly for LRSB	2 nos	
	Plastic refractory	10 tons	
	Chicken wire mesh	200 Mts	
	Gland packing for valves of all types, sizes.		
	Gland Packing for Soot blowers		
	Gaskets for valves		
	Ropes of various types and sizes		

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	Pre- Cut Fins for Water wall, SCW	1000 mtr.	Higher quantity require for Water Wall
	All types of electrodes for welding	300 Kgs	Higher quantity required for Water Wall and Economiser area depending on site condition
	Filler wires for welding	100 Kgs each	
	Jointing & sealing compounds		
	Cold welding compounds		
	Rust removers		
	Argon gas		
	DA gas		
	Oxygen gas		
	Pressure seal gasket/ Soft iron gasket for each type, size of high-pressure valves	2 Nos.	

S.No.	Description	Unit	Suggestive Stock
APH Drive Systems			
	GEAR BOX ASSY(Speed reducer) FOR APH	No	1
	PINION AND GEAR FOR APH GEAR Box	Set	2
	FREE WHEEL COUPLING/Over running clutch	No	2
	Fluid Coupling	No	2
	Multi Disc Coupling	No	2
	BIBBY COUPLING/DRIVE COUPLING	No	2
	Bearings for the APH Gear Box	Set	2
	TAPER LOCK BUSH FOR DRIVE GEAR BOX	No	2
	Rack and Pin assy	Set	1
	Pinion for main drive	No	1
	AIR MOTOR	No	2
	air motor silencer	No	2
	Lubricator	No	4
	Oil Seals for Gear Box	No	4
APH Guide Bearing			
	Guide Bearing	No	2
	Adapter Sleeve for APH GB	No	2
	GUIDE BEARING HOUSING	No	1
	SEAL TUBE FOR AIR PRE HEATER	No	2
	CENTRAL TRUNION SHAFT FOR APH	No	2
	Trunion Shaft Bolts	Set	2
Support Bearing			

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	Support Bearing	No	2
	SUPPORT BEARING HOUSING	No	1
	BEARING RUNNER PLATE	No	1
	Body		
	HE RADIAL SECTOR PLATE	No	2
	CE RADIAL SECTOR PLATE	No	2
	AXIAL SEAL PLATE FOR AIR PREHEATER	No	2
	COLD END ROTOR T-Bar FOR APH	Set	2
	Hot End T-bar for APH	Set	2
	BYPASS SEAL ANGLE FOR APH	Set	2
	PIN RACK ASSY FOR APH	Set	1
	STATIC SEAL PLATE-HE SECT.PLATES- PAPH	Set	2
	Water washing deluge pipe with Nozzles	Set	1
	Soot blower Nozzles	Set	1
	LUB OIL SYSTEM OF APH		
	LUBE OIL PUMP FOR APH SB	No	1
	LUBE OIL PUMP FOR APH GB	No	1
	PRESSURE RELIEF VALVE OF LUBE OIL SYSTEM	No	2
	COMPLETE LUB OIL FILTER ASSEMBLY	No	4
	LOVEJOY FLEXIBLE COUPLING	No	4
	LUB-OIL COOLER assy	No	2
	SET OF GASKETS AND O RING	No	2
	Basket for APH		
	Baskets for APH as per rolling plan	Set	1
	Seals for APH		
	HE and CE Radial seals	Set	Full Population
	HE and CE Bypass seals	Set	Full Population
	Axial Seals	Set	Full Population

Ducts (500MW)

S.No.	Description	Unit	Suggestive Stock
	Ducts		
	SAIL Hard plates	MT	100
	ERW pipes(80/100/125)	Mtr	100
	Dampers	Set	1
	Damper Bush	Set	1
	Damper Actuator Gear Box	Set	1
	Seals for isolating gates(Hast alloy seals)	Set	2
	Fabric material of different thickness and material for NMEJ	RM	100
	Refractory for high wear prone area	MT	10
	Wear plate for High wear prone area	MT	10
	Expansion Joints Hot PA and Cross-over ducts	Sets	5

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Fans (500MW)

S.No.	Description	Unit	Suggestive Stock
Axial PA Fans			
	Bearing Housing ASSY	No	1
	Blade for PA Fan	Set	1
	Bearings for Fan Housing	Set	1
	Servomotor Assy	No	1
	Corten shaft and Coupling	Set	1
	Oil Seals for Bearing Housing	Set	2
	Oil Seals for Servomotor Assy	Set	2
	End rod bearing for Blade Pitch mechanism	Set	2
	Lub Oil Pump	No	1
	Lub Oil Filter	No	2
Axial FD Fans			
	Bearing Housing	No	1
	Blade for PA Fan	Set	1
	Bearings for Fan Housing	Set	1
	Servomotor Assy	No	1
	Corten shaft and Coupling	Set	1
	Oil Seals for Bearing Housing	Set	2
	Oil Seals for Servomotor Assy	Set	2
	End rod bearing for Blade Pitch mechanism	Set	2
	Lub Oil Pump	No	1
	Lub Oil Filter	No	2
Radial ID Fans			
	Impeller assy	Set	1
	Impeller Shaft assy	Set	1
	Beariing Housing	No	1
	Journal bearings for ID fan	Set	2
	ID fan coupling between fan and motor	set	2
	Lub Oil Pump	No	1
	Lub Oil Filter	No	2
	Lube oil cooler assy for ID fan -	set	2
Seal Air Fans			
	Seal air fan impeller with shaft -	No	1
	Seal air fan bearing -	Set	1

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Mills (500MW)

SI No	Description	UO M	Qty
1	Grinding Roll	NO	6
2	Bull Ring Segment	SET	2
3	Bull Ring Clamping Ring	SET	2
4	Bowl Hub Cover	NO	2
5	Airport Ring Assembly	SET	2
6	Inner Cone	NO	1
7	venturi Collar	NO	2
8	Outlet Venturi	NO	2
9	MPO Plate	NO	2
10	MPO	NO	8
11	MDV	NO	8
12	Orifice (complete repl in One Unit)	SET	1
13	Inverted Cone	NO	2
14	Classifier Shaft & blade	NO	10
15	Scraper Shoe	NO	10
16	Scraper guard	NO	10
17	Lower Skirt	SET	2
18	Insulation cover plate	SET	1
19	RJ bearing upper	NO	6
20	RJ bearing Lower	NO	6
21	Worm Shaft	NO	1
22	Worm Shaft bearing (Radial)	NO	2
23	Worm Shaft bearing (Thrust)	NO	2
24	Mill Coupling	NO	2
25	Center Feed pipe (Lower)	NO	2
26	Center Feed Pipe (Upper)	NO	2
27	Center pipe above coupling	MTR	20
28	MOE	NO	14
29	FIE	NO	8
30	PF Straight pipe	MTR	50
31	Different types of PF bends		20 % of pop
32	Victaulic coupling	NO	8
33	Victaulic coupling Gasket	NO	50
34	Feeder Belt	NO	6
35	feeder Head Pulley	NO	1
36	Feeder tail Pulley	NO	1
37	Feeder Span Roller	NO	12
38	Feeder Center Pulley	NO	1
39	Feeder pulley bearings	SET	25 % of pop
40	Feeder Main gear box spares	SET	1
41	Feeder Main gear box bearing	SET	1
42	COC gear box Assembly	NO	1
43	COC assembly	NO	1

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44	HAG, CAG seals		50% of Pop
45	HAD, CAD seals		50% of Pop
46	Plates of HAG	NO	1
47	Flap of HAD dampers	NO	4
48	Expansion joints near HAG & CAG	NO	2
49	Feeder outlet chute	NO	2
50	Bunker outlet Gate	NO	1
51	Feeder inlet - outlet gate	NO	1
52	Mill Lub Oil Pump assembly	NO	1

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Annexure – VIII Suggestive List of Overhauling Spares-Turbine

A. Turbine (200/500MW KWU)

System	Items	Quantity	Remarks	MACHINE RATING
HPT	MS pipe/casing U-seal ring as per Drawing	2		210/500 MW
	MS pipe /casing U-seal ring-Unmachined	2		210/500 MW
	Inner/Outer casing U-seal ring as per Drawing	1		210/500 MW
	Inner-Outer casing U-seal ring -Unmachined	1		210/500 MW
	Inner/Outer I-seal ring-Unmachined	1	Pool item	210/500 MW
	HP I/L side gland U-seal ring-Machined	2	For ver-2 onwards	500 MW
	HP Exhaust side gland bush U-seal ring-Machined	1		500 MW
	HPT exhaust Pipe/Casing U-seal ring Unmachined	1		210/500 MW
	HPT exhaust Pipe/Casing U-seal ring as per Drawing	1		210/500 MW
	HPT exhaust Pipe/Casing serrated gasket	2		210 MW
	CRH NRV flap, hinge pin assembly	1 set		210 / 500 MW
	Breach Nut & inlet inserts	1 set	Pool item	210/ 500 MW
	Rotaing Blades	1 Set	Pool item	210/ 500 MW
	Rotor		Pool spares	210/ 500 MW
	Stationary Blades		Pool item	210/ 500 MW
	Module		Pool spares	210/ 500 MW
	Gland seal segments (front) with springs	1 Set		210/ 500 MW
	Gland seal segments (rear) with springs	1 Set		210/ 500 MW
	Exhaust side gland bush	1	For ver-2 onwards	500 MW
	Inner casing parting plane studs & nuts	1 Set		210/ 500 MW
	Inner casing to outer casing radial keys	1 Set		210/ 500 MW

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	Inlet side gland cover studs	1 Set	For ver-2 onwards	500 MW
	High density balancing weight	30		210/ 500 MW
	CRH flange studs	10%		210/ 500 MW
	HP outer casing inserts with one set of Un machined angle ring & threaded ring	1 each		210/ 500 MW
	Stud and Cap Nut for forced cooling dummy flange	1 set		210/ 500 MW
	HP rotor fins with caulking wire for balance drum and gland seal area	1 set		210/ 500 MW
	Evacuation valve stem, disc and seat (if applicable)	1		210/ 500 MW
IPT	Inlet pipe angle ring-Unmachined	1 set		210/500 MW
	Inlet pipe threaded ring-Unmachined	1 set		210/ 500 MW
	Extraction Pipe Angle ring-Unmachined	1 set		210/500 MW
	Extraction Pipe threaded ring-Unmachined	1 set		210/ 500 MW
	Rotating Blades	1 Set	Pool item	210 / 500 MW
	Rotor	1	Pool spares	210 / 500 MW
	Stationary Blades		Pool item	210 / 500 MW
	Inner/Outer Casing-Module		Pool spares	210 / 500 MW
	Inner casing parting plane studs & nuts	1 Set		210 / 500 MW
	Outer casing studs	10%		210 / 500 MW
	Inner casing holding down bolts, palm packers, circular washer, axial keys & Radial Keys etc.	2 Set		210 / 500 MW
	Inlet U-seal ring, if applicable	1		500 MW
	High density balancing weight	30		210 / 500 MW
	IP I/L, extraction & exhaust flange studs	10 % of each type		210 / 500 MW
	Stud and Cap Nut for forced cooling dummy flange	1set		210 / 500 MW
	IP rotor fins with caulking wire for gland seal area	1 set		210 / 500 MW

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	Gland seal segments with springs (both front & rear)	1 Set		210 / 500 MW
LPT	Extraction-1 bellows	1 Set		210 / 500 MW
	Extraction-2 bellows	1 Set		210 / 500 MW
	Extraction-3 bellows	1 Set		210 / 500 MW
	LPT inlet pipe bellows	1 Set		210 / 500 MW
	LPT casing to gland box bellows	1		210 / 500 MW
	Clamping pieces for free standing blades	1 Set		210 / 500 MW
	locking plates for free standing blades	1 Set		210 / 500 MW
	Free standing Rotating Blades	1 Set	Pool item	210 / 500 MW
	Rotor	1	Pool spares	210 / 500 MW
	Stationary Blades	1 Set	Pool item	210 / 500 MW
	LPT inner Inner casing parting plane stud & cap nut	1 set		210 / 500 MW
	LPT Inner outer casing and outer casing parting plane fasteners	10% of each type		210 / 500 MW
	Fasteners of LPT inner outer casing manhole	20%		210 / 500 MW
	Palm packers, axial keys & radial keys	2 nos each		210 / 500 MW
	High density balancing weight	30		210 / 500 MW
	Lead diaphragm	10 No.		210 / 500 MW
Bearing	Gland seal segments with springs	1 Set		210 / 500 MW
	Gland Housing (Front & rear)	1 set	Pool item	210 / 500 MW
	Brg. 1 babbited shell with matching support	1 Set		210 / 500 MW
	Brg. 2 babbited shell with matching support and thrust pads	1 Set		210 / 500 MW
	Brg. 2 liner	1		210 / 500 MW
	Brg. 3 babbited shell with matching support	1		210 / 500 MW
	Brg. 4 babbited shell with matching support	1		210 / 500 MW
	Brg 5 babbited shell and spherical support pads	1		210 / 500 MW
	Brg 6 babbited shell and spherical support pads	1		210 / 500 MW
	Brg. 7 babbited shell with matching support (if applicable)	1		210 / 500 MW

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	Pedestal anchor bolts	1 of each type		210 / 500 MW
	Pedestal oil guard fins	1 set of all pedestal		210 / 500 MW
	MOP front & rear bearings	2 set		210 / 500 MW
Jacking Oil system	Jacking oil flexible hoses for each bearing	1 Set		210 / 500 MW
	Jacking oil pump assembly	1		210 / 500 MW
	Jacking oil to each bearing NRV	1 Set		210 / 500 MW
	JOP individual relief v/v	2		210 / 500 MW
	JOP start up vent v/v	1		210 / 500 MW
	JOP filter elements	4		210 / 500 MW
	Jacking oil header pressure regulating valve assembly	1		210 / 500 MW
Control Oil system and Governing Rack	Hydraulic Governor bellows	1		210 / 500 MW
	Bellow for LP Bypass control rack assy	1		210 / 500 MW
	Main Trip Valve assembly	1		210 / 500 MW
	All Kinds of "O" rings & diaphragm	1 Set		210 / 500 MW
	Control fluid pump complete assy.	1		500 MW
	Control fluid accumulator bladders	min. 10%		500 MW
	CF pump discharge NRV	1 of each type		500 MW
	Critical rubber item for Governing system/Valves (Auto groove ring set, Omega sealing ring, Compact sealing ring, Automatic gasket Set etc	2 sets		210 / 500 MW
	Filter elements	1 Set		210 / 500 MW
Lubricating Oil System	Main Oil pump assembly complete with Bearings	1 Set		210 / 500 MW
	HPT/MOP gear Coupling	1 Set		210 / 500 MW
	MOP suction Oil Injector assembly	1	Pool item	210 / 500 MW
	Oil strainer/filter elements	2 sets		210 / 500 MW
	AOP assembly	1		210 / 500 MW
	EOP assembly	1		210 / 500 MW
	HP/IP (1 set normal + 1 set oversize)	1 Set each		210 / 500 MW

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Coupling Bolts	IP/LP (1 set normal + 1 set oversize)	1 Set each		210 / 500 MW
	LP/Gen (1 set normal + 1 set oversize)	1 Set each		210 / 500 MW
	Generator-Exciter coupling studs with bushes	1 Set each		
Turning gear	Stationery Blades/ Nozzle	1 Set	Pool item	210 / 500 MW
	Rotating blades	1 Set	Pool item	210 / 500 MW
HPSV/CV	Servomotor complete assy.	1 of each type	Pool item	
	Valve cone, stem, threaded ring of cone, bush, pin	1 each		210 / 500 MW
	Valve body threaded ring	1 No.		210 / 500 MW
	Stop & CV body stud & nuts	10%		210 / 500 MW
	Valve U seal ring	1 each		210 / 500 MW
	Stop valve both test valve liner with Margin	4 No.		210 / 500 MW
	Stop valve pilot of valve block assembly	1 No.		210 / 500 MW
	Valve cover bush	1 each		210 / 500 MW
	Helical & disc springs of servomotors	1 set		210 / 500 MW
	Coupling assy.	1 each		210 / 500 MW
	Control valve servomotor : Precontrol & main control valve pilot & liner	1 set		
	Stop valve servomotor springs/Piston/Bottom Plate	1 set		210 / 500 MW
IPSV/CV	Servomotor complete assy.	1 of each type	Pool item	210 / 500 MW
	Valve cone, stem, threaded ring of cone, bush, pin	1 each		210 / 500 MW
	Stop & CV body stud & nuts	10%		210 / 500 MW
	Helical & disc springs of servomotors	1 set		210 / 500 MW
	Coupling assy.	1 each		210 / 500 MW
	Valve cover bush	1 each		210 / 500 MW
	Helical & disc springs of servomotors	1 set		210 / 500 MW
	Stop valve servomotor Piston/Bottom Plate	1 set		210 / 500 MW
	Control valve servomotor : Precontrol & main control valve pilot & liner	1 set	Pool item	
	Servomotor complete assy.	1	Pool item	210 / 500 MW

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LPBP SV/CV	Stop valve stem	1		210 / 500 MW
	Stop valve cone	1		210 / 500 MW
	LPBP spray pressure switch	1		210 / 500 MW
	control valve stem & cone	1		210 / 500 MW
	Coupling assy.	1 each		
HPBP	Pressure control valve stem	2 Set		210 / 500 MW
	Pressure control valve seat	2 Set		210 / 500 MW
	valve cage	1 set		210 / 500 MW
	spray valve stem & seat	2 Set		210 / 500 MW
MS Strainer	MS strainer Gasket	1 Population		
	Element along blanking device gasket, protection ring with Margin	1		210 / 500 MW
HRH Strainer	HRH strainer Gasket	1 Population		
	Element along blanking device gasket, protection ring with Margin	1		210 / 500 MW
Condense r	CW Pipe to Condenser flexible joint assembly	1 each		210 / 500 MW
	Condenser inlet/outlet CW valve rubber cords (valve door seal)	2 sets each	To be stored in AC Room	210 / 500 MW
	Zinc anode, if applicable	10%		210 / 500 MW
	COLTCS balls	5000 per unit		210 / 500 MW
	CW inlet, outlet BFV gear box assembly	1 No each type		210 / 500 MW
	CW interconnection valve rubber cords (valve door seals)	1 sets	To be stored in AC Room	210 / 500 MW
Evacuatio n System	Vacuum pump assembly	1 No.		210 / 500 MW
	Vacuum pump rotating assembly (Rotor & impeller)	1 No.		210 / 500 MW
	Vacuum pump rotating assembly (Comprising of both stage impeller)	1 No.	For Edwards Model	500 MW
	Vacuum Pump Mech seal assy, Bearing set, Circulation pump assembly	1 No.		210 / 500 MW
	Vacuum pump coupling assy with coupling bushes/tyre coupling	1 No.		210 / 500 MW
	Vacuum pump inlet valve & NRV assembly	1 No.		210 / 500 MW

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	Cooler element	1 No.	Most applicable for sea water application, Others can maintain as per requirement	210 / 500 MW
Heaters	LP heater-1 safety valve shell side	1 No.		210 / 500 MW
	LP heater-2 safety valve shell side	1 No.		210 / 500 MW
	LP heater-3 safety valve shell side	1 No.		210 / 500 MW
	HP heater-5 safety valve shell side	1 No.		210 / 500 MW
	HP heater-6 safety valve shell side	1 No.		210 / 500 MW
	Deaerator safety valve	1 No.		
	Deaerator spray nozzles	30%		210 / 500 MW
	Deaerator tray assembly	10%		
	Feed water system (I/L & O/L iso. v/v, BFP discharge valve, FRS station valve) stem, disc, pressure seal, actuator bush etc.	1 set		210 / 500 MW
Boiler Feed Pump	BFP cartridge assembly	1/3rd of Population		210 / 500 MW
	Booster pump rotating Assy	1 No.		210 / 500 MW
	Booster pump casing assembly	1 No.		
	Mechanical seals for booster pump & BFP	2 sets		210 / 500 MW
	All bearings for booster pump & Main pump	1 set		210 / 500 MW
	All types of wear rings for booster pump & Main pump	1 set		
	Discharge valve stem and disc & pressure seal	1 set		210 / 500 MW
	Recirculation CV trim set	2 sets		210 / 500 MW
	SH & RH spray isolation valve, spare pressure seal gasket	1 set		210 / 500 MW
	SH & RH spray NRV	1 set each		210 / 500 MW
	BFPs one set of flexible coupling, spare coupling disk plate (element assembly), set of fasteners	1 each type)		210 / 500 MW

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	Booster & BFP suction strainer	1 No. each		
	Bearings set of hydraulic coupling	1 set		
	Pump Insert of Lub oil & working oil pump of hydraulic coupling	1 No. Each		
	Hydraulic coupling assembly	1	Pool item	210 / 500 MW
Condensate system	CEP assembly	1		210 / 500 MW
	One set of CEP cutless rubber bearings, Shaft sleeves, impeller, shaft & coupling, wear ring, throttle bush	1 set		210 / 500 MW
	CEP Coupling (Pump to Motor)	1 No.		
	CEP Suction strainer, suction compensator, suction safety relief valve	1		210 / 500 MW
	CEP discharge valve stem and disc, discharge NRV,	1		210 / 500 MW
	CEP thrust cum journal bearing & mechanical seal	1 each		210 / 500 MW
	Condensate line NRV before Deaerator assembly	1		210 / 500 MW
Turbine gland seal steam system	Gland steam condenser assembly	1	Pool item	210 / 500 MW
	Gland steam control valve trim set	1		210 / 500 MW
TDBPF Drive Turbine	DT rotor assy. With emergency governor, if applicable	1		500 MW
	Set of Bearings including thrust bearing	1		500 MW
	AOP, EOP & JOP complete assy.	1 No. each		500 MW
	JOP PRV	1 No.		500 MW
	Overspeed cum trip device assy.	1 No.		500 MW
	Front & rear gland assembly, Oil guard rings	1 set		500 MW
	SV, MCV, ACV with servomotor	1 set		500 MW
	balancing weight, pedestal disk springs	1 set		500 MW
	Gear box complete assembly	1 type each		
	TDBFP inlet steam strainer	1 No.		500 MW
Generator	High density balancing weight for generator &	30 nos each		210 / 500 MW

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	exciter rotor			
ACW ECW system	Pump assembly	1 no each type		210 / 500 MW
	All B/F V/V gaskets (BFV door seal)	2 no each type		210 / 500 MW
	Pump discharge NRV, spare NRV disk/seat	1 each		210 / 500 MW
	PHE Plate, end plate, set of gaskets	10%		210 / 500 MW
	all BFVs gear box assembly	1 each		210 / 500 MW
	Motor/Pump coupling, Mech seal	1 no each type		210 / 500 MW
Primary Water System	Primary Water Pump assy.	1		210 / 500 MW
	Discharge NRV	1		210 / 500 MW
	Mechanical seal assy.	1		210 / 500 MW
Seal Oil System	Seal oil pump assembly (Air side)	1		210 / 500 MW
	Seal oil pump assembly (H2 side)	1		210 / 500 MW
	Seal oil relief v/v assy.	1		210 / 500 MW
High energy drain v/v & Other control valves	Set of valve trim & gaskets for all type of control valves used in TG area	1 set for each type		210 / 500 MW
	One set of High energy drain valves	1 of each type		210 / 500 MW
	drain valves (1/2", 1", 1.5" & 2": CI 800 to 3500 class, A105/F11/F22)	10%		210 / 500 MW
	One set of pressure seal gaskets/ bonnet gaskets/ gland packings for all type of high pressure valves in TG area	1 set		210 / 500 MW

NOTE: Items listed as "Pool item" may not be mandatorily kept available at site before OH. The availability of the same may be identified in other station in consultation with OS-ST in case of any urgent requirement after dismantling. However, the item must be available at site prior to OH in case it has been planned to be replaced as per scope of work of OH. 2. Items have been categorized depending upon the equipment to be overhauled. Therefore, items can be selected depending upon planned scope of work for particular overhauling.

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Critical Unit assemblies to be maintained at site for Turbine & aux. for ensuring system availability.

Sl. No.	DEPT	NAME OF THE EQPT/ UNIT	POPULATION	NAME OF THE SPARE	No of Spares to be available at site.
1	TMD	TG		TG Bearing No.1	1 no.
2	TMD	TG		TG Bearing No.2	1 no.
3	TMD	TG		TG Bearing No.3	1 no.
4	TMD	TG		TG Bearing No.4	1 no.
5	TMD	TG		TG Bearing No.5	1 no.
6	TMD	TG		TG Bearing No.6	1 no.
7	TMD	TG		TG Bearing No.7	1 no.
8	TMD	TG		TG Bearing No.8 (if envisaged)	1 no.
9	TMD	TG		Main turbine MOP assembly	1 no.
10	TMD	TG		Main turbine MOP bearings	1 no. each type
11	TMD	TG		Main Turbine AOP complete assembly	1 no.
12	TMD	TG		Main Turbine EOP complete assembly	1 no.
13	TMD	TG		Main Turbine JOP complete assembly	1 no.
14	TMD	TG		Main Turbine JOP PRV, Relief v/v & startup vent v/v	1 no. each type
15	TMD	TG		Control fluid pump complete assembly	1 no.
16	TMD	TG		Control fluid accumulator bladders	3 nos. each type
17	TMD	TG		Control fluid pump Discharge NRV	1 set
18	TMD	TG		Strainer's gaskets	2 sets
19	TMD	TG		Lead diaphragm	5 nos.
20	TMD	TG		MOP coupling	1 no.
21	TMD	GENERATOR		Primary Water Pump Assembly	1 no.
22	TMD	GENERATOR		Seal oil pump assembly (Air side)	1 no.
23	TMD	GENERATOR		Seal oil pump assembly (H2 side)	1 no.
24	TMD	TDBFP		TDBFP drive Turbine rotor	1 no.
25	TMD	TDBFP		BFP Main pump cartridge	1 no.
26	TMD	TDBFP		Booster Pump assembly	1 no.
27	TMD	TDBFP		AC lub oil pump for TDBFP	1 no.
28	TMD	TDBFP		EOP for TDBFP	1 no.
29	TMD	TDBFP		TDBFP JOP Complete assembly	1 no.

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30	TMD	TDBFP		Turbine to booster pump gear box	1 no.
31	TMD	CEP		CEP Cartridge	1 no.
32	TMD	MISC.		Vacuum pump assembly	1 no.
33	TMD	MISC.		vacuum pump suction NRV	1 set
34	TMD	MISC.		Voith coupling complete assy. or one set of bearings, AOP & WOP	1 Set
35	TMD	MISC.		DMCW/ECW pump	1 no. each type
36	TMD	MISC.		ARCW pump	1 no.
37	TMD	MISC.		Compensators	1 no. each type
38	TMD	MISC.		HPBP stem & seat	1 no. each type

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Annexure – IX - Minimum spares & Assemblies for C&I Systems

1. Spares for Control system/DDCMIS/PLC (covering FSSS /Turbine /OLCS / CLCS including Gas turbines):

SL. NO	Material Description	Quantity Recommended.	Status
1.	Electronic modules of each type and model for control system, PLC, Data communication system, Master and Slave Clock etc. (This shall include all type of cards like I/O cards, Remote I/O cards, communication/Interface cards, controller cards, CPU module or Card, logic cards, Bus coupler/Interface hardware / other communication devices etc.).	10 % or min 1 No of each type	
2.	Power Supply cards / Modules & Power Packs for control system and UPS / Charger.	10 % or min 1 No of each type	
3.	Prefab interconnecting cables with connectors.	10 % or min 1 No of each type	
4.	System bus cable with connectors.	10 % or min 1 No of each type	
5.	I/O bus cable with connectors for remote I/O units.	10 % or min 1 No of each type	
6.	Servers for different functions with operating system and software loaded complete with all accessories.	1 no of each type	
7.	Fuses of different ratings for UPS, DC power supplies and electronic cards.	100%	

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2. C&I spares for Main Turbine, TD BFP/MDBFP, HP Bypass and LPBypass:

Sl. NO.	Material Description	Quantity available	Status
1.	Electro-Hydraulic Converter coil/Servo Unit / Proportional valve and position feedback transmitter for main turbine, HPBP and LPBP.	1 no of each type	
2.	Position feedback transmitter for main turbine HP CV /IPCV.	1 no of each type	
3.	Main Turbine trip solenoid / complete trip device.	1 no of each type	
4.	TSE Thermocouple.	10 % or min 1 No of each type	
5.	TSI sensor for axial shift, expansions, speed and vibration.	10 % or min 1 No of each type	
6.	Electronic Cards/PCB's/interface units/Load Transducers/Frequency Transducer etc. of all types used in Turbo supervisory system /Turbine control.	10 % or min 1 No of each type	
7.	Electro-Hydraulic Converter coil/Servo Unit/ proportional valve and position feedback transmitter for TDBFP turbine.	10 % or min 1 No of each type	
8.	TDBFP Turbine trip solenoid / complete trip device.	Min.1 no of each type	
9.	RTD / Thermocouple / E-P converter / Air Lock Relay	Min. 3 Nos. of each type	
10.	Servo valve, Blocking Unit, Oil line hoses, Feedback transmitter, Filter for HPBP / LPBP.	Min.1 no of each type	
11.	Auto Synchronizer.	Min.1 no. of each type and model.	

3. C&I spares for Boiler, CLCS & OLCS:

Sl. NO.	Material Description	Quantity available	Status
1.	Solenoid valve for Critical applications like- HOTV / LOTV / HORV/ Spray Block valves etc.	10 % or min 1 No of each type	
2.	Pneumatic actuator/ Power Cylinder for critical applications.	1 No of each type	
3.	Diaphragm/seal kit for pneumatic actuator.	10 % or min 1 No of each type	
4.	Electric regulating actuator including control valves, Starting device, Speeder Gear, blade pitch and vanes etc.	1 no of each type	

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5.	Positioner for critical control valves/actuators like FRS, BFP/CEP Recirculation valves and ID/FD/PA blade pitch and IGV.	10 % or min 1 No of each type	
6.	Scoop actuator motor with brake.	1 no of each type	
7.	Scoop actuator gear box.	1 no of each type	
8.	Scoop feedback transmitter.	1 no of each type	
9.	Controller module for scoop tube actuator.	1 no of each type	
10..	Transmitters used in protection/Interlock circuits of Boiler,Turbine and major auxiliaries.	10 % or min 1 No of each type	
11.	Switches used in protection/Interlock circuits of Boiler, Turbine and major auxiliaries.	10 % or min 1 No of each type	
12.	Scanner & Ignitor assembly.	10 % or min 2 No of each type	
13.	Oxygen probe with electronic module complete / any other critical analyzers/spares.	Min.1 No of each type.	
14.	Boiler/Turbine Desk/Draw out console EPB.	Min.1 No of each type.	

4. Control cables

Sl. NO.	Material Description	Quantity available	
1.	Instrumentation cables of various types including shielded, twisted pair cables for analog & binary signals, thermocouple extension cables, heat resistant Teflon insulated cables, FibreOptic cable, Co-ax cable.	10 % or min 1 Km of each type	

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Annexure – X - BHEL Generator overhauling material list (500 MW)

Sl no	Assembly y name	Material description	Drg no	Qty	UoM
1	General assembly	O-RING 193X6 (OIL PIPELINE)	139370100000017	6	NO
2		O RING 52X3 (PRE-CHAMBER LVL DETECTOR TO END SLD SEALING)	139370100000018	15	NO
3		O-RING 115X4 (EE END SHIELD MANHOLE COVER)	139370100000050	3	NO
4		SCREW HEX A-8.8 M16X35 (EE END SHIELD MANHOLE COVER)	139370100000051	20	NO
5		SCRU CAP SOC TH HD A-8.8	139370100000022	10	NO
6		SCRU HEX A -8.8 M20X60	139370100000023		NO
7		SCRU CAP SOC TH HD A-8.8 M16X50	139370100000024	10	NO
8		WASHER	139370100000036	30	NO
9		SCRU CAP SOC A12.9 M16X70	139370100000021	20	NO
10		SCRU CAP SOC A12.9 M36X110 (END SHIELD)	139370100000020	10	NO
11		BRG SADDLE FIXING BOLT SLEEVE	913936010010035	4	NO
12		BRG SIDE SUPPORT FIXING BOLT SLEEVE	913936010010005	4	NO
13		INSULATING PIN 91393601001 ITEM 44	913936010010044	2	NO
14		WASHER LOCK WITH LUG -ST 8.5	139120101700043	30	NO
15		INSULATING PLATE	013912010170041	2	NO
16		Locking ring	0139370100000042	5	NO
17		Insulating washer	0139370100000041	20	NO
18		SLEEVE 24X20.5, L=67	9139360100100 35	10	NO
19		Insulation strip	913936010010009	2	NO
20		NIPPLE (Tite seal pushing)	139120101700070	10	NO
21		STATOR END SHIELD SOCKET CAP 91391201003	139120101700011	5	NO
22		STATOR END SHIELD SOCKET CAP 91391201003		5	NO
23		10 mm cord		100	M
24		8 mm cord		50	M
25		GEN DRG KWU-, O-RING-297*4 MAT.FLUOR (Coupling cord)		4	NO
26		GASKET 01391201017 ITEM 54 (outer oil catcher)	139120101700054	4	NO
27		GLASS TEXOLITE SLEVE L-22, OD-17, ID-12MM (Outer oil catcher)		20	NO
28		TITE SEAL,	139120101700085	20	KG
29		SCRU CAP SOC A12.9 PM30X90	139120101700046	20	NO
30		WASHER TAB A-21 ST	139120101700007	20	NO
31		BOLT HEX B8.8, Brg Parting plane bolt	9139020100000010	2	NO

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32		WASHER DISC SPRING, PN.9139020100000008	9139020100000000	10	NO
33		TAPER PIN. P.NO.9139020100000013	9139020100000013	1	NO
34		LOCK WASHER M16	13907010000005	100	NO
35		WASHER 17 ST	13907010000007	50	NO
36		INSUL WASHER	13907010000008	30	NO
37		SLEEVE, 20X16.5-30 .01390701000 PT- 9	13907010000009	30	NO
38		INSULATING SEGMENT.NO.01390701000 PT-10	13907010000010	10	NO
39		INSULATING SEGMENT 01390701000 PT12	13907010000012	10	NO
40	Compres sor assembly	LOCK WASH 8X12	13907010000016	50	NO
41		WASH LOC LUG 13	13907010010005	50	NO
42		WASH LOC LUG 13	13907010010008	30	NO
43		TUNGUED LOCK WASHER 39	9139070100003	10	NO
44		THREADED BUSH	9139070100005	3	NO
45		HEX NUT M39x3	9139070100006	3	NO
46		BOLT HEX M16X100	9139070100008	30	NO
47		SCRU HEX PM8X16	9139070100009	10	NO
48		LOCK WASHER M8	9139070100010	100	NO
49		RUB PROFILE 105-002 IT012,9-139-07-01000	9139070100012	8	NO
50	Gas cooler assembly	H2 SEALING ROD PACKING RINGS A57F87D39		20	NO
51		U-GASKET FOR H2 COOLERS		8	NO
52		H2 SEALING ORING 265X8 A00A26F83	139120101700053	10	NO
53		5 MM Champion water gasket		100	KG
54		M12 lock washer		200	NO
55	Seal assembly	SEAL RING TE#11393501003	1-154-01-0100001-0.(ver -01)	1	NO
56		SEAL RING EXC END #11393501003	1-154-01-0100000 (ver -00)	1	NO
57		HIGH PRESSURE HOSE 1391701000C	139170100001	4	NO
58		GEN DRG 0-139-35-01001-28, SEALING RING	139350100100028	10	NO
59		O-RING-32*3	139350100100029	10	NO
60		LOCK WASHER, (Seal body bolt (M24) lock washers)	139340100100013	10	NO
61		VITON Rubber profile 7.5x9 D cord	139340100100008	30	M
62		INSULATION GASKET (Seal body Gasket without hole)	139350100100007	2	NO
63		INSULATION GASKET (Seal body Gasket with hole)	139350100100009	2	NO
64		NSULATION SLEEVE, (Seal body bolts)	139350100100010	10	NO
65		INSULATING WASHER (Seal body bolt (M24) washers)	139350100100011	20	NO
66		LOCK WASHER (Seal body bolt (M24) lock washers near Seal body PP),	139340100100018	5	NO

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67		INSULATION SLEEVE (Seal body sleeve Seal body PP)	139350100100015	6	NO
68	Inner oil catcher assembly	Lock WASHER A10-5-ST003 1-139-35-01001	113935010010003	50	NO
69		INSULG WASR 008/1-139-35-01001	113935010010008	20	NO
70		INSULG SLV I009/1-139-35-01001	113935010010009	10	NO
71		INSULATION GASKET IT010,1-139-35-01001	113935010010010	2	NO
72		GEN DRG 1-139-35-01001-11, INSU GASKET	113935010010011	2	NO
73		SEALING HALF RING, 0139120100000004	0139120100000004	100	NO
74	Outer oil catcher assembly	INSULATING SLEEVE 17x2.5x22	139120101700013	10	NO
75		INSULATING WASHER 01391201017 ITEM 16	139120101700016	20	NO
76		GASKET 01391201017 ITEM 54	139120101700054	4	NO
77	Air gap seal assembly	BOLT M10x70	139410100000002	4	NO
78		WASHER LOCK WITH LUG-BR 10.5	139410100000003	30	NO
79		SCRU HEX A 04CR18NI10. M12X40	139410100000005	8	NO
80		WASHER TABA-BR 10.5	139410100000006	20	NO
81		SCRU HEX A 04CR18NI10. M10X70	139410100000008	6	NO
82		WASHER	139410100000010	20	NO
83		WASHER TABA-BR 13.0	139410100000012	20	NO
84	PW System	O' RING 26.3X2.4	139120100800004	16	NO
85		GASKET PN40	139120100800006	10	NO
86		O' RING 50X4 „0139120100800020		10	NO
87		O' RING 64.2X5.7 „0139120100800022		4	NO
88		GASKET PN40 ITEM NO 21 OF PW TANK ASSY.	153020100300021	10	NO
89		GASKET PN40	915006010010045	6	NO
90		GASKET PN40	915006010010046	10	NO
91		GASKET PN40	915006010010047	6	NO
92		GASKET PN 40 OF DRG NO 91500601001-94	915006010010094	4	NO
93	Consumables	Sillicon sealant RTV-32		20	KG
94		3 mm neoprene rubber gasket sheet		50	KG
95		5 mm neoprene rubber gasket sheet		100	KG
96		3 mm style 59 oil gasket		100	KG
97		CRC Lectra clean/ equivalent		20	Ltr
98		Bectol Red insulating varnish		20	KG
99		Bectol Grey insulating varnish		10	Kg
100		Suitable thinner of varnish		10	Kg
101		Black rubber paint cooler water		10	Kg
102		Dendrite		10	Kg
103		Loctite 211			

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Annexure – XI - Suggestive Spare List of Ash Handling System (500 MW)

S.NO.	DESCRIPTION	UOM	QTY	Remarks
1	CASTABLE REFRACTORY	MT	60MT	30MT FOR 210 MW
2.	HYDRO EJECTOR NOZZLE	NOS	4	100% REPLACEMENT
3	HYDRO EJECTOR THROAT PIECE	NOS	4	100% REPLACEMENT
4	HYDRO EJECTOR WEIR PIECE	NOS	4	100% REPLACEMENT
5	CLINCKER GRINDER ROLLER/SHAFT /SHAFT SLEEVE	SET	4	100% REPLACEMENT
6	CLINCKER GRINDER WEAR PLATE	NOS	8	100% REPLACEMENT
7	CLINCKER GRINDER BEARING	NOS	16	CONDIIION BASED
8	CLINCKER GRINDER BEARING HOUSING	NOS	16	CONDITION BASED
9	CLINCKER LANTERN RING	NOS	16	100% REPLACEMENT
10	CLINCKER GRINDER BEARING RING	NOS	16	100% REPLACEMENT
11	BEARING SLEEVE	NOS	16	CONDITION BASED
12.	INLET PIECE	NOS	4	
13	NOZZLE INLET	NOS	4	
14	INCLINED FEED GATE ROLLER/WEDGE	NOS	24	CONDITION BASED
15	ROLLER SHAFT /BRACKET	NOS	24	CONDITION BASED
16	SS WEAR STRIP FOR GATE	NOS	4	100% REPLACEMENT
17	SS WEAR STRIP FOR GATE HOUSING	NOS	4	100% REPLACEMENT
18	POWER CYLINDER SEAL KIT	SET	4	100% REPLACEMENT
19	POWER CYLINDER PISTON	NOS	4	CONDITION BASED
20	POWER CYLINDER SHAFT	NOS	4	CONDITION BASED
21	POWER CYLINDER CYLINDER	NOS	4	CONDITION BASED
22	REFRACTORY COOLER HEADER	NOS	18 HEADERS	CONDITION BASED

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23	HP FLUSHING HEADER/HP FLUSHINGNOZZLE	NOS	8 HEADERS	CONDITION BASED
24	SEAL TROUGH FLUSHING NOZZLE	NOS	18	CONDITION BASED
25	SEAL TROUGH DRAIN VALVE	NOS	4	CONDITION BASED
26	ECO FLUSHING APPARATUS NOZZLE	NOS	8	100% REPLACEMENT
27	BUFFER HOPPER BAG FILTER	SET	8	100% REPLACEMENT
28	ROTARY SEGREGATING VALVE SEAT	NOS	4(FOR ONE BUFFER HOPPER)	RSV SEAT AND GATE SHOULD REPLACE IN ALL BUFFER HOPPERS
29	ROTARY SEGREGATING VALVE GATE	NOS	4(FOR ONE BUFFER HOPPER)	
30	PRESSURE EQUALISING VALVE SEAT	NOS	4(FOR ONE BUFFER HOPPER)	
31	PRESSURE EQUALISING VALVE DISC	NOS	2(FOR ONE BUFFER HOPPER)	PEV SEAT & DISC SHOULD BE REPLACE IN ALL BUFFER HOPPERS
32	PEV BODY	NOS	2(FOR ONE BUFFER HOPPER)	
33	FLUIDISING STONE	NOS	AS PER DESIGN	100% REPLACEMENT
34	ASH INTAKE VALVE OF BUFFER HOPPER	NOS	1	CONDITION BASED
35	SEGREGATING VALVE OF VACUUM HEADER	NOS	AS PER DESIGN	CONDITION BASED
36	LATERAL OF VACUUM HEADER	NOS	AS PER DESIGN	CONDITION BASED
37	MATERIAL HANDLING VALVE BODY L/R	NOS	AS PER DESIGN	CONDITION BASED
38	VALVE SEAT /SLIDE OF MHV	NOS	AS PER DESIGN	CONDITION BASED
39	COUPLING GASKET OF VACUUM HEADER/FLY ASH TRANSFER LINES	NOS	AS PER SITE REQUIREMENT	CONDITION BASED
40	BENDS/LATERAL	NOS	AS PER SITE REQUIREMENT	CONDITION BASED
41	VACUUM PUMP ASSEMBLY	NOS		CONDITION BASED
42	BUFFER HOPPER FLUDIZING BLOWER	NOS		CONDITION BASED

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Annexure – XII - Minimum Critical spares For Overhauling Work of CW System

S. No.	Description
C W Pumps	
01	C W Pump Impeller One No. of Each Model
02	CW Pump Shaft Sets One No. of Each Model
03	Thrust Bearing One No. of Each Size of Various Models
04	Line Bearing Set One Set for Each Model
05	Bowl Assembly One Assy. For each Model
06	Actuator Assembly for Butterfly Valves One Assembly of Each Model
07	Complete Power Pack of Electrohydraulic Actuator of CWP Discharge Valve One Assy. For Each Model.
Cooling Towers	
01	Blade Assembly One Set for Each Models
02	Gear Box Assembly One for Each Model
03	Shaft Assembly One for Each Model
04	Riser Valve Assembly One for Each Model
05	Riser Valve Actuator Assembly One for Each Model
06	Coupling Assembly One for Each Model
07	Gear Box Bearing Set One Set for Each Model
08	Gear Set One Set of Each Model